

# Air Brakes (Structures) [25-26]

## Purpose

Flight scoring is based on the demonstrated altitude achieved relative to target apogee and successful recovery. The accuracy of the launch vehicle's actual apogee is worth 70% of the point value of flight performance, and so trajectory planning is important. One strategy in reducing the error between target apogee and flown apogee is the implementation of an Airbrake system.

The goal of the Airbrake is to lower apogee by removing kinetic energy via drag inducing elements. The deployment occurs in the coast phase of the flight. Airbrakes are an active control system whose outputs are the acutation drag inducing elements and inputs are trajectory (accelerometer, IMU) data.

$$Points = 350 - \left( \frac{350}{0.3 \times Apogee_{Target}} \right) \times |Apogee_{Target} - Apogee_{Actual}|$$

*where Apogee<sub>Target</sub> may equal either 10,000 ft AGL or 30,000 ft AGL*

## Competition Requirements

A table of requirements for the system (What requirements is this system governed by?) Think about all the parts of the rocket that your system involves. These should all come directly from the [System Requirements Document](#)

| Requirement Index | Requirement Explanation                                                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| COMP 7.1.1        | All launch vehicle active flight control systems shall be implemented strictly for pitch and/or roll stability augmentation, or for aerodynamic braking       |
| COMP 7.1.2        | Under no circumstances will a launch vehicle entered in the IREC be actively guided towards a designated spatial target.                                      |
| COMP 7.1.4        | Neutral state is defined as one which does not apply any moments to the launch vehicle (e.g., aerodynamic surfaces trimmed or retracted, gas jets off, etc.). |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>COMP 7.2.1</b>  | Launch vehicles implementing active flight controls shall be naturally stable without those controls being implemented (e.g., the launch vehicle may be flown with the control actuator system (CAS) – including any control surfaces – either removed or rendered inert and mechanically neutral, without becoming unstable during ascent).                                                     |
| <b>COMP 7.3.1</b>  | Control actuator systems (CAS) shall default to a neutral state whenever either an abort signal is received for any reason, primary system power is lost, or the launch vehicle’s attitude exceeds 30° from its launch elevation.                                                                                                                                                                |
| <b>COMP 7.4.1</b>  | Control actuator systems (CAS) shall remain in a neutral state until:<br><br>OR 7.4.1.1 The launch vehicle’s boost phase has ended (i.e., all propulsive stages have ceased producing thrust).<br><br>OR 7.4.1.2 The launch vehicle has crossed the point of maximum aerodynamic pressure (max Q) in its trajectory.<br><br>OR 7.4.1.3.2 For 10K flights: an altitude of 2,000 m (6,500 ft) AGL. |
| <b>COMP 8.2.1</b>  | Launch vehicles shall be constructed to withstand the operating stresses and retain structural integrity under the conditions encountered during handling and transportation and during rocket flight.                                                                                                                                                                                           |
| <b>COMP 10.3.1</b> | Launch vehicles shall maintain a dynamic stability margin of at least 1.5 body calibers, regardless of Cg movement and/or shifting center of pressure Cp location, from launch through the first recovery system deployment event.(Ascent Stability)                                                                                                                                             |
| <b>COMP 10.4.1</b> | Launch vehicles shall not be “over-stable” during their ascent, defined as having a static stability margin >4 calibers or a dynamic stability margin during flight >6 calibers.                                                                                                                                                                                                                 |

## Internal Requirements

This should be similar to Competition Requirements, except that these are requirements you (the designer) write to outline for yourself. These can be useful in further defining your design

| <b>Requirement Index</b> | <b>Requirement Explanation</b>                                                                                            |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>SYS.1</b>             | All components on the LV that interface with the airframe shall have at least .005" clearance with the ID of the airframe |
| <b>SYS.4</b>             | Structural components shall be designed to a minimum MoS of 0.5                                                           |

|                 |                                                                                                                                                                                                        |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>STRUC 2</b>  | Launch vehicles shall be constructed to withstand the operating stresses and retain structural integrity under the conditions encountered during handling as well as rocket flight.                    |
| <b>STRUC 6</b>  | Airframe joints which implement "coupling tubes" shall be designed such that the coupling tube extends no less than one body caliber on either side of the joint – measured from the separation plane. |
| STRUC 7         | All airframe joints shall be stiff to prevent bending by more than 1°.                                                                                                                                 |
| <b>STRUC 20</b> | FEA shall be conducted for all structural components of the LV.                                                                                                                                        |
| STRUCT 23       | Material trade analysis shall be conducted for all structural components                                                                                                                               |
| <b>STRUC 36</b> | All components shall be qualified by test to 1.25x limit load.                                                                                                                                         |
| <b>STRUC XX</b> | The Airbrake mounts shall constrain the assembly during launch and recovery.                                                                                                                           |
| <b>STRUC XX</b> | The Airbrake mount shall fully withstand the 1.5x maximum induced drag force output, and no greater than 5x.                                                                                           |
| STRUCT 21       | Manual stress calculations shall be conducted for all structural components of the LV.                                                                                                                 |
| STRUCT X.X.X    | Drag Inducing Elements must not create a net torque about the vehicle CG                                                                                                                               |
| STRUCT X.X.X    | All Drag Inducing Elements must be mechanically linked                                                                                                                                                 |
| STRUCT X.X.X    | Airbrakes aperture should not cause airframe to violate structural integrity.                                                                                                                          |
| STRUCT X.X.X    | Rocket must maintain stability throughout airframe deployment                                                                                                                                          |

## WEEK OF 9/1/25:

### PDR Goals:

- Identify relevant Competition Requirements
- Start outlining Internal Requirements
- Ideate designs for trade study
  - What trade offs are there to consider
  - Surface Area
  - Torque needed
  - Redundancy
  - Mass

- Determine who is responsible for what system. Airbrake Mechanism vs Airbrake Integration.
- Update Page Requirements

There are two primary approaches to increasing vehicle drag.

- The first being a set of "leaves" that deploy flat rectangular surfaces.
- Or a hinged deployment that applies a torque.

There are many other design approaches (hinged, piston, pulley etc...)

We need to start characterizing the performance we expect this system to have, and then based on that start evaluating design parameters.

We need to determine predicted airspeed so we can start getting order of magnitude estimations.

Determine Material

What wind tunnels are available?

## **WEEK OF 9/8/25**

In order to evaluate different designs for our trade study, we need to characterize the environment we expect deploy the Air Brakes. The important conditions to consider are the wind speed, altitude, and air density. Using this information along with a dynamic model of our rocket we can start approximating the vehicle dynamics during the coasting phase. Finally we need to determine a maximum change in altitude as a system parameter. With Air Brakes the best approach is to design a rocket that overshoots the target apogee. The altitude change required by our system can be approximated by taking the overshoot value, adding uncertainty and multiplying by a factor of safety. Then the model can be used to predict how Air Brakes of different cross sectional area and drag coefficients operate. For preliminary analysis we will not take into account active control, but rather model Air Brakes as a system that transitions from the neutral state to an active state where it is fully deployed and does not actuate further during flight.

## **CHARACTERIZING DEPLOYMENT CONDITIONS:**

Air Brakes can be deployed if the following conditions are met:

(7.4.1.1) The launch vehicle's boost phase has ended (i.e., all propulsive stages have ceased producing thrust).

(7.4.1.2) The launch vehicle has crossed the point of maximum aerodynamic pressure (max Q) in its trajectory.

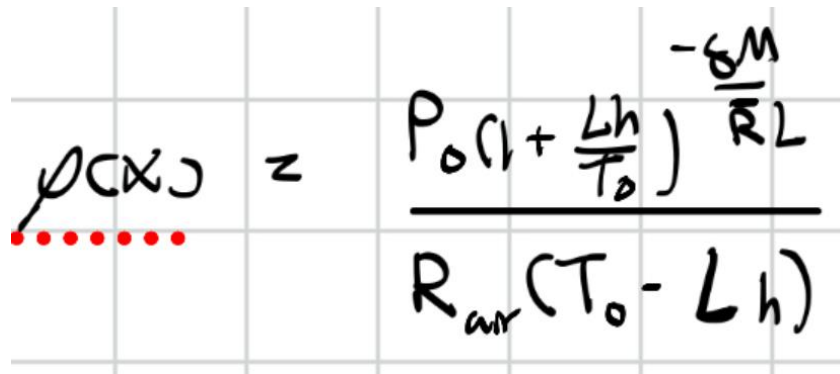
(7.4.1.3.2) For 10K flights: an altitude of 2,000 m (6,500 ft) AGL.

We will use a previous Andromeda flight data to determine the dynamic state of the LV when these conditions are met.

### Approximating Air Density as a function of altitude:

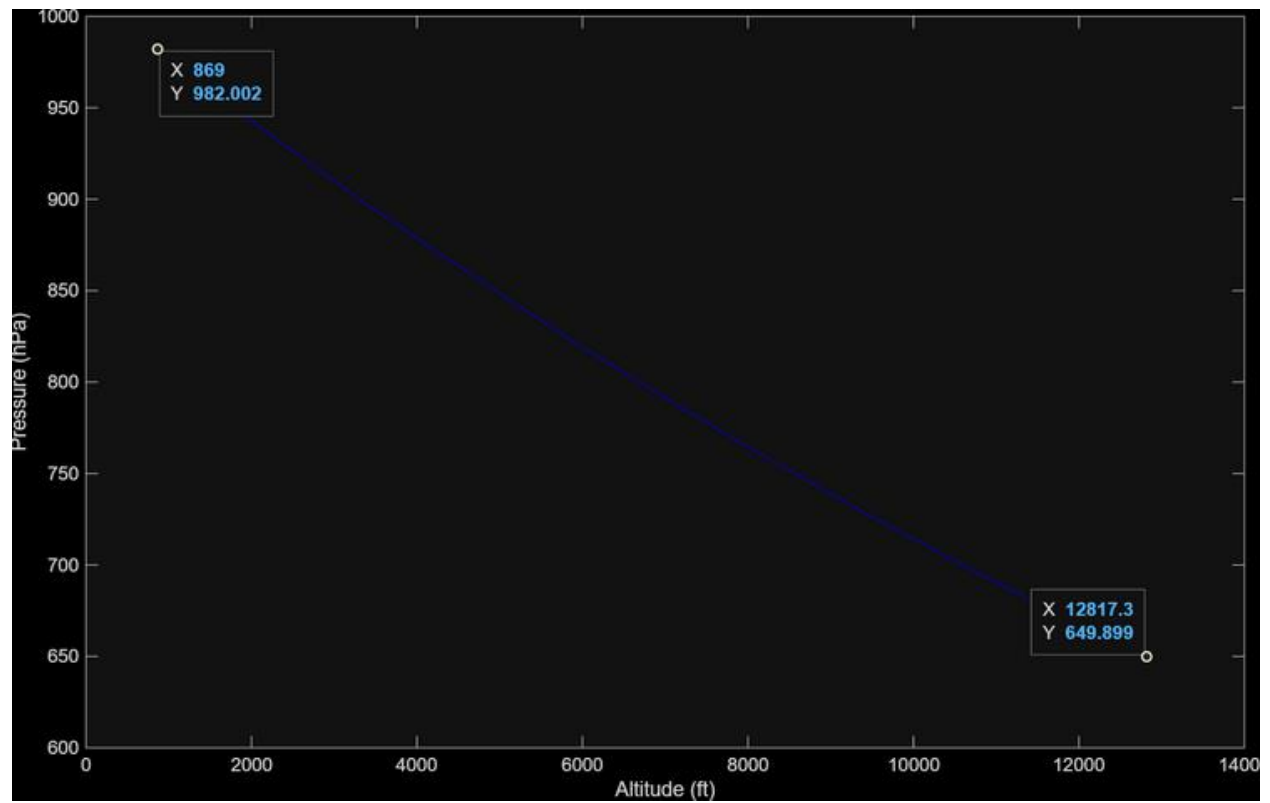
In both the Test Flight Aerodynamic Pressure estimation and in the numerical integration test cases air density is approximated in the following way:

- Pressure is approximated using the barometric formula for linearly decreasing temperature
- Density is derived by treating air as an ideal gas with specific gas constant  $\sim 287$  J/kgK

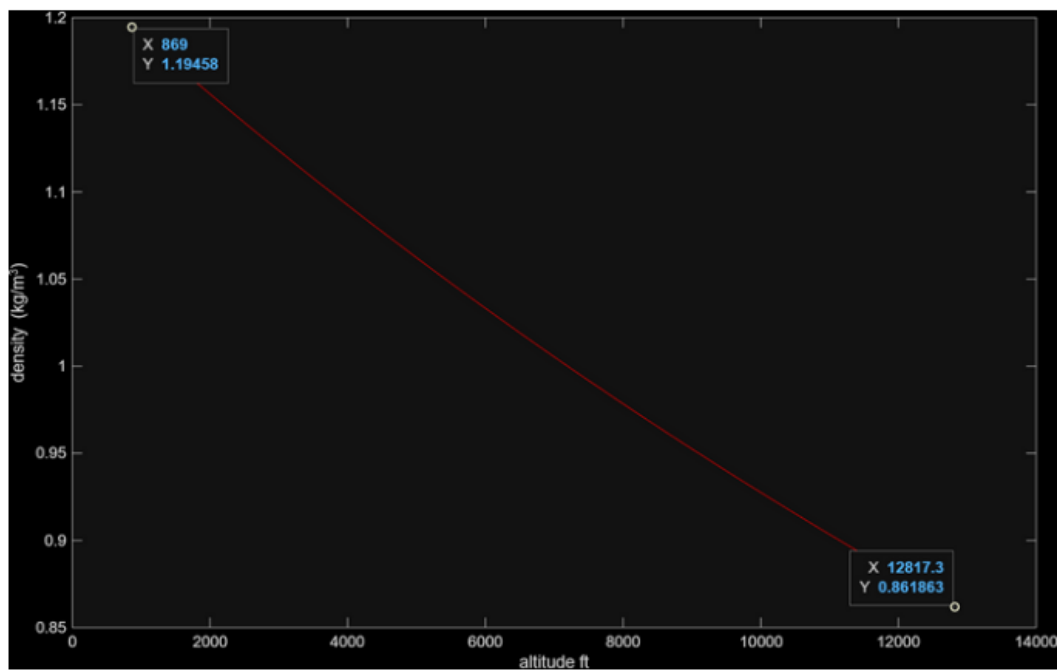


The image shows a handwritten equation on a grid background. On the left, the text 'Pressure' is written above a horizontal line of red dots. To the right of the dots is an equals sign. Further right is the formula: 
$$P = \frac{P_0 \left(1 + \frac{Lh}{T_0}\right)^{\frac{-gM}{R}}}{R_{air}(T_0 - Lh)}$$

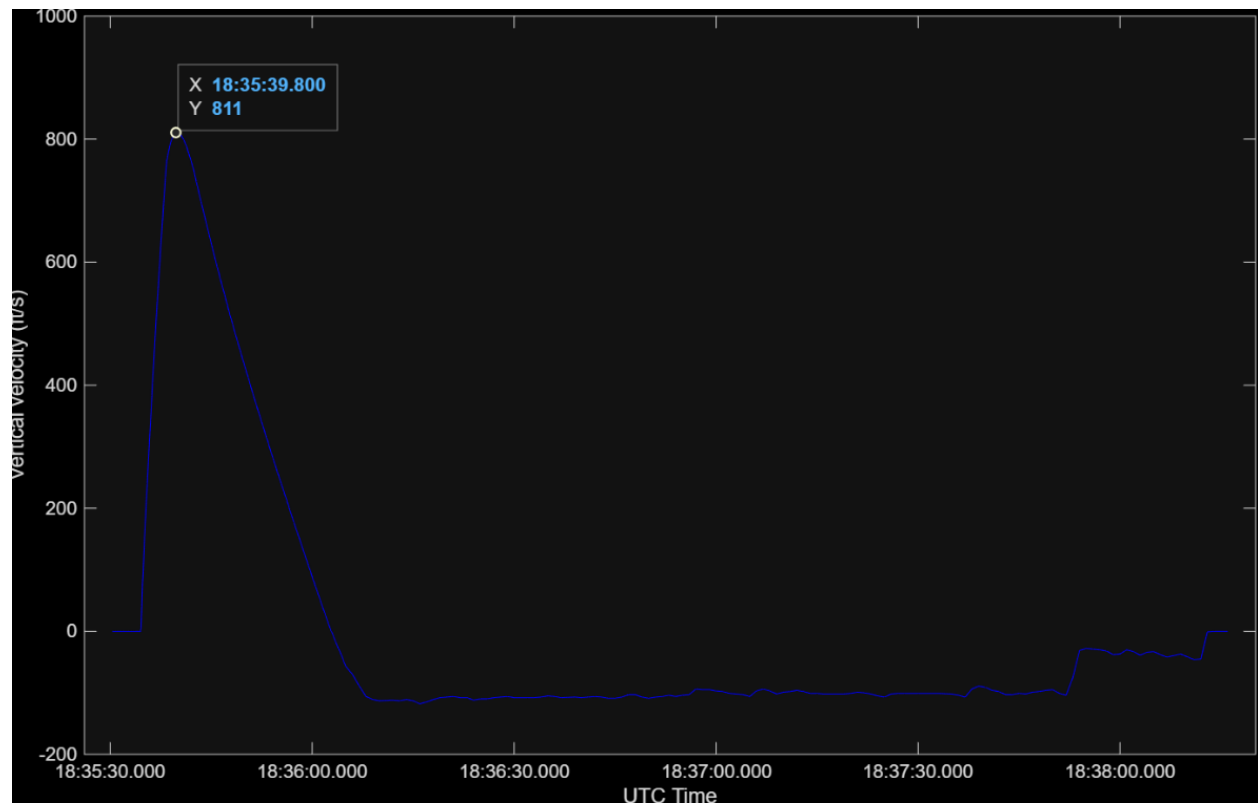
Pressure versus Altitude:



Density versus Altitude:



Velocity versus Time



Max Velocity: **811 ft/s** (Mach 0.72) - t +4.9s

Aerodynamic Pressure vs Time



## Formulation of Q

Max Q = 33.708 kPa -t +4.9s



Launch t +0s (18:35:34.9)

Max Q: t +4.9s (18:35:39.8) 3640 ft, 811 ft/s

**Coast Phase(MAV Close): t+7.8s (18:35:42.8) 5990 ft 731 ft/s**

Fail Safe: t+8.9s (18:35:44.5 )+ 6500 ft

Now with these deployment conditions characterized we can do a comparison of the pros and cons of each condition. Earlier deployment conditions allow for longer duration (potentially on the order of 3 seconds) of additional time to effect drag upon the LV. Increasing the duration of the deployment time can reduce the magnitude of the force required to realize the same impulse. However the Max-Q condition could potentially be dangerous, as drag force magnitudes are greatest at this point. Furthermore the trajectories at this initial condition are more difficult to model. Our model assumes there is not thrust forces from the motor. MAV Close is a software



defined time after launch when the MAV closes off the run tank from the motor/nozzle. While the majority of thrust is lost after the burn end time ( $\sim t+4.4s$ ) which occurs before MAV close, there could still be gases being released from the LV. MAV Close is the best definition for the start of the vehicle coast phase. The fail safe altitude was reached around 2 seconds after MAV Close. Waiting until the rocket reaches 6500 ft AGL further reduces the time to effect drag, and further decreases in the magnitude of the drag force we can expect to realize, without adding any modeling benefits. For this reason Boost Phase Conclusion/Coast Phase start is the middle deployment time for our airbrakes.

| Condition | Maximum Aerodynamic Pressure (Max-Q)                                                                                                                                      | Boost Phase Conclusion (MAV Close)                                                                                                  | Reaching Altitude of 6500 ft AGL.                                                   |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Occurs    | T + 4.9s                                                                                                                                                                  | T + 7.8s                                                                                                                            | T + 9.8s                                                                            |
| Pros      | <ul style="list-style-type: none"> <li>- Earliest deployment condition</li> <li>- Dense Atmosphere</li> </ul>                                                             | <ul style="list-style-type: none"> <li>- <b>Easier to model forces</b></li> <li>- <b>Easy to determine during flight</b></li> </ul> | <ul style="list-style-type: none"> <li>- Easy to determine during flight</li> </ul> |
| Cons      | <ul style="list-style-type: none"> <li>- Difficult to model drag coefficients approaching transonic Mach Numbers</li> <li>- Harder to determine during flight.</li> </ul> | <ul style="list-style-type: none"> <li>- <b>Less time to effect drag on LV</b></li> </ul>                                           | <ul style="list-style-type: none"> <li>- Less time to effect drag on LV.</li> </ul> |

## 1D - DYNAMIC MODEL

### Assumptions:

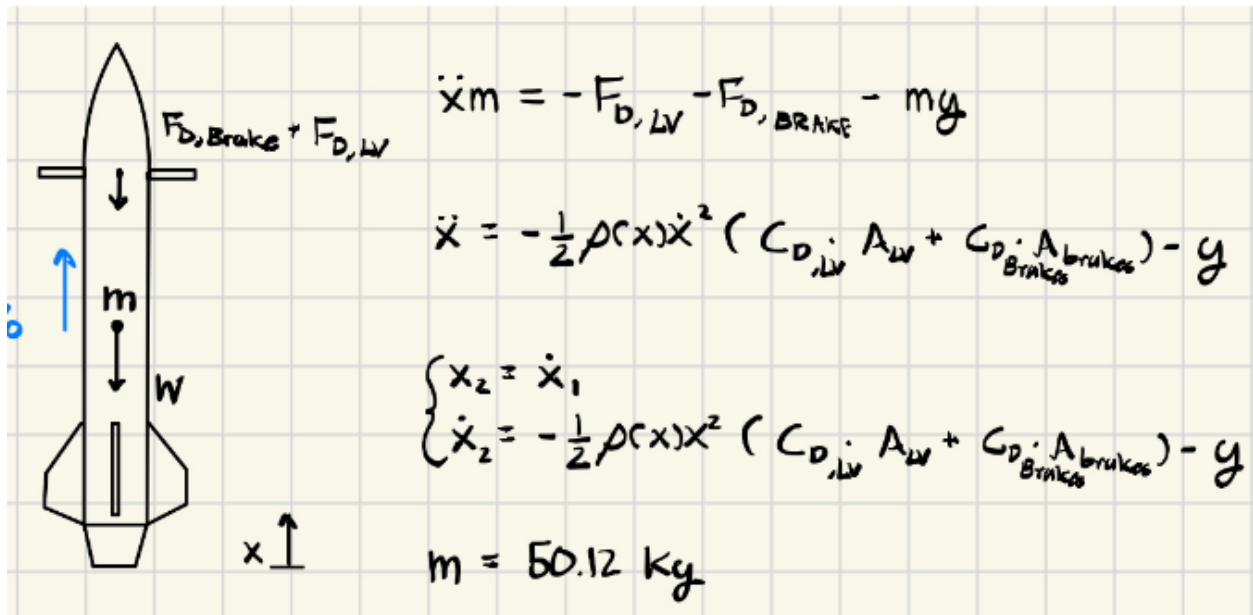
Only force acting on rocket is gravity and aerodynamic drag (neglecting air-frame friction)

Motion of the Rocket is Purely Vertical (1D)

Pressure and Temperature are given from Barometric Equation, where Temperature decreases according to average environmental lapse of  $-6.5K/km$ .

Treating Dry Air as an ideal gas with specific gas constant  $\sim 287 J/kgK$ .

Superposition of Drag Forces:



Where the mass is the dry mass of Andromeda: 110.5 lb or **50.12kg**

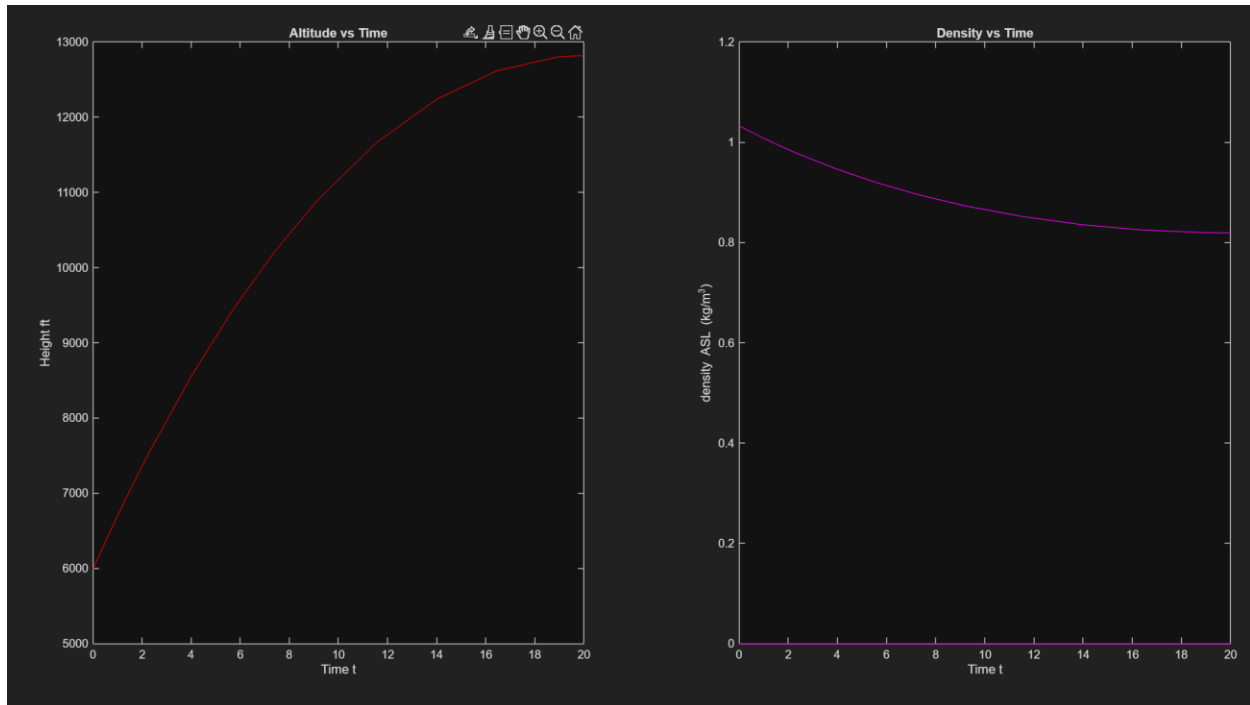
In order to approximate the Cd of the rocket, we use the initial conditions from start of the coast phase( $t+7.88s$ )

$z_0 = [5990 \text{ ft}, 731 \text{ ft/s}]$

The reference area of the rocket traveling vertical upwards is  $\pi \cdot 9 \text{ in}^2$  or **0.002026 m<sup>2</sup>**.

We set the cross sectional area of the air brakes to zero, and approximate the drag coefficient of Andromeda in our model by determining what Cd results in a maximum altitude ASL of 12817ft after numerical integration

**Cd = 0.5158**



We can get a sense of model accuracy by noting that during test flight the duration between MAV close (18:35:42.8) and Apogee (18:36:03) is **20.1s** where as the model predicts a duration **20.0s**.

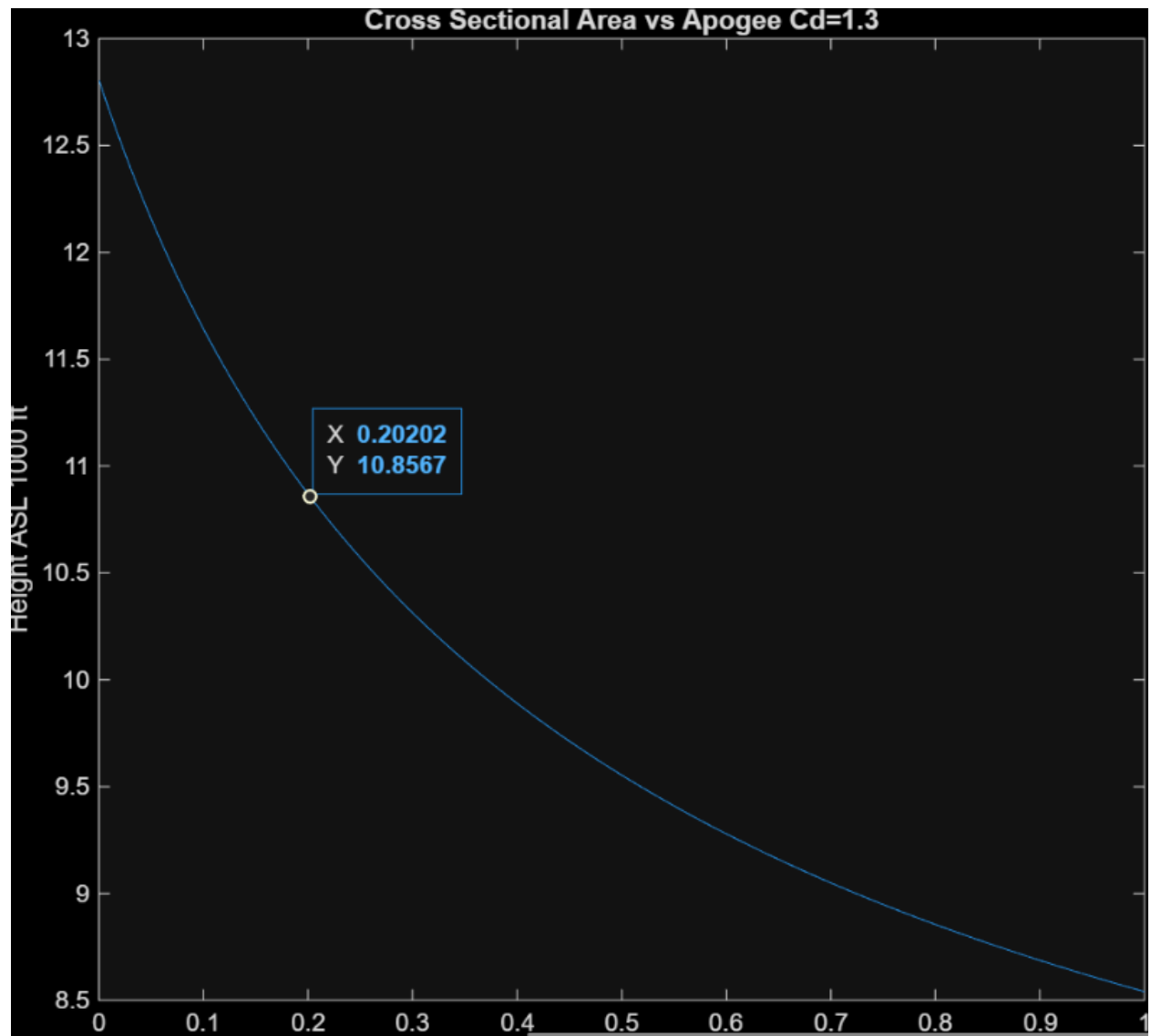
## ESTIMATING AIRBRAKE PERFORMANCE:

The effectiveness of an Airbrake in reducing the final height depends on their cross-sectional area, drag coefficient, and air density through deployment.

Many air-brakes utilize blunt drag inducing elements. The drag coefficient for a thin flat surface like a plate, perpendicular to flow is around 1.3 so we will use that value for the drag coefficient of our airbrakes.

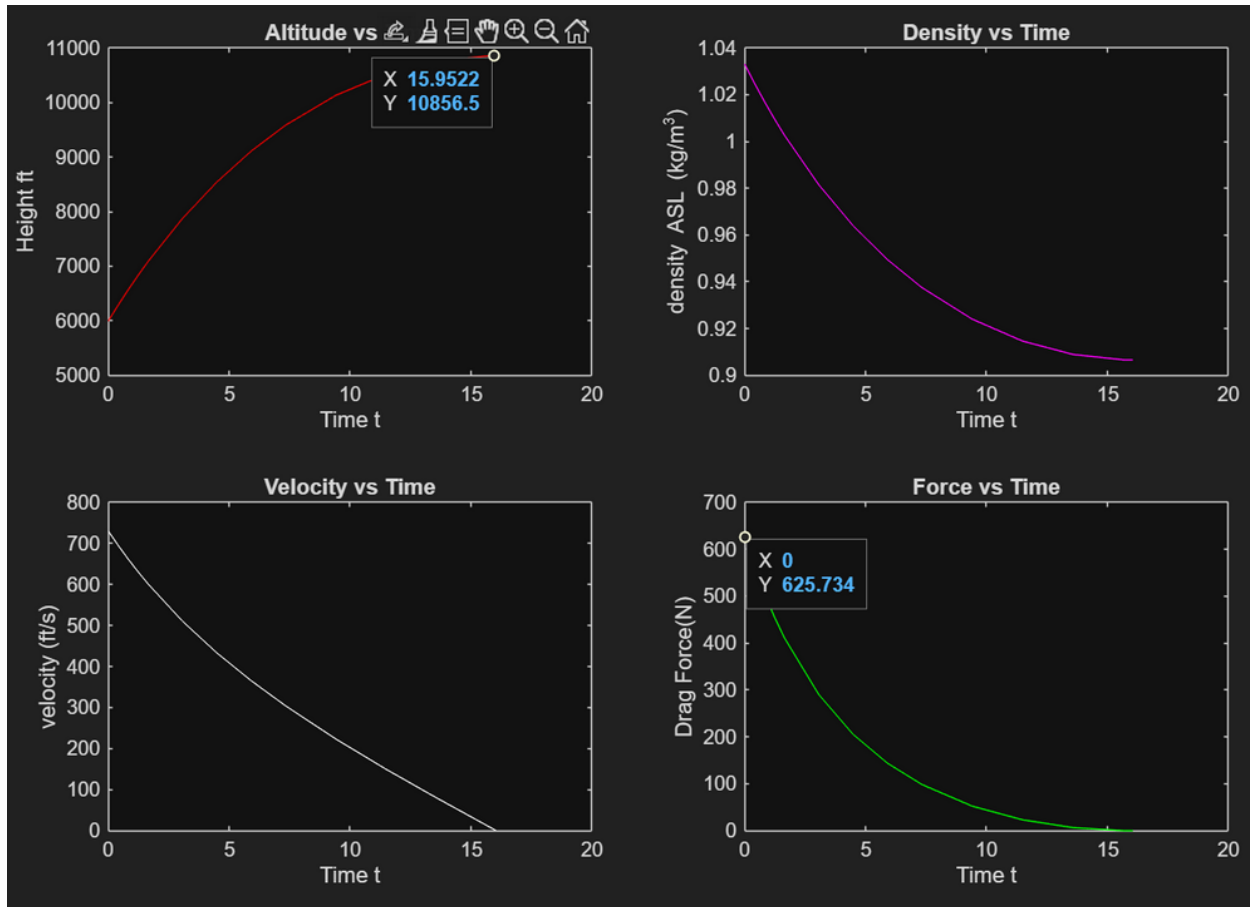
For this stage in our design a drag coefficient need only be accurate within 20%. For your fluid environment (Mach 0.64 and High Reynolds number) you can approximate the drag coefficient for bluff bodies to be constant and equal 1.3

We can use the numerical integrator to explore what the apogee of the rocket would be as a function of the cross sectional area of the airbrakes



So from this graph we can determine **.202 ft<sup>2</sup> with Cd = 1.3** would have resulted in a final altitude AGL of 10k ft.

Here is the performance of **.232 ft<sup>2</sup>** of airbrakes with a Cd of **1.3** deployed at **5990 ft** with an initial velocity of **731 ft/s**



It takes **15.96s** for the airbrakes to halt of LV at an altitude of **~10k AGL**, It experiences at peak drag force of **625.734N** at deployment.

Caveat: it seems this model has a sharp decrease in altitude for initial conditions prior to MAV close. While this may in part be due to weird oxidizer stuff not currently part of the model we suspect there also may be a limitation in the way treat the drag coefficient as being independent of Mach Number. So for earlier stages of deployment around Max-Q their may be more to consider in characterizing the model.

## WEEK OF 9/21/25

### Section 0: Effects of Temperature Variation.

Temperature is modeled as  $T_0$  - Lh. Ground Level: 860 (266.25m) ft ASL.

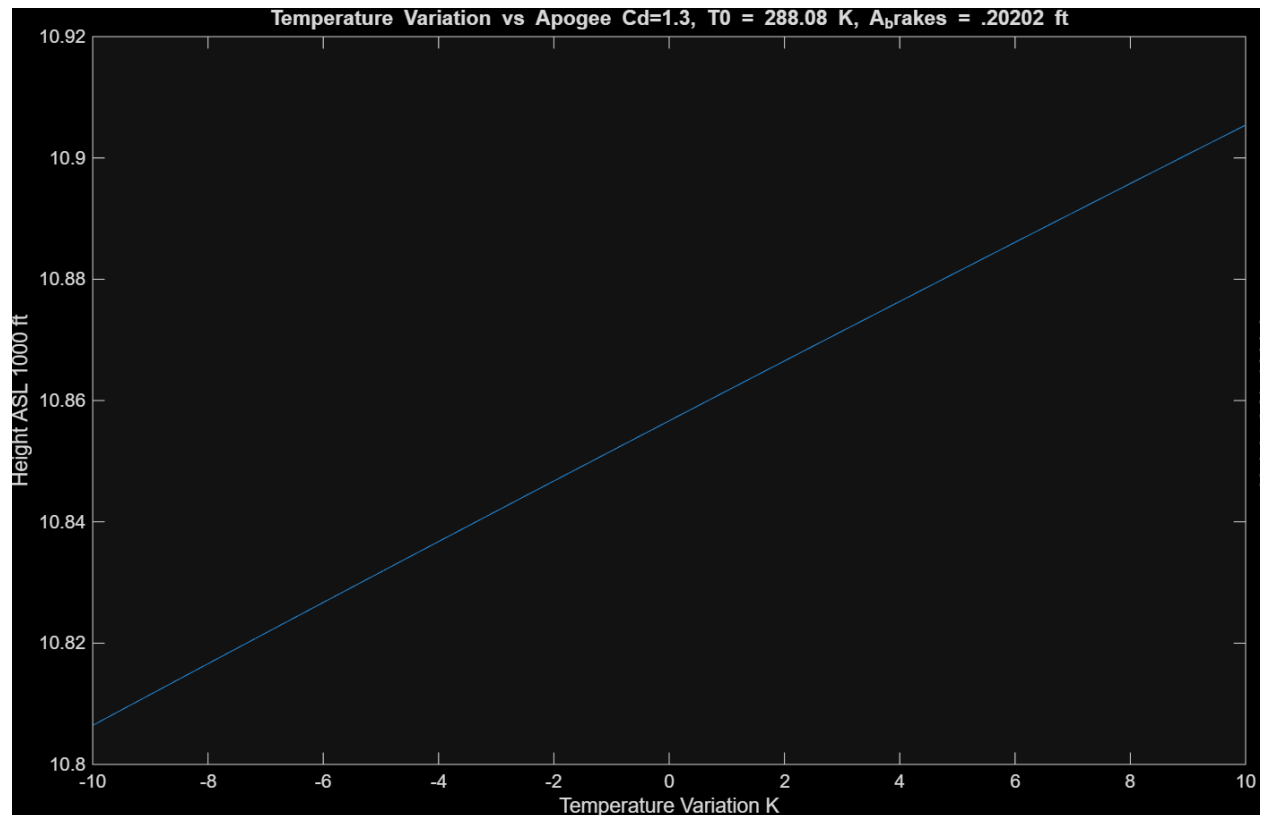
$$T_0 = 288.15\text{K} (15^\circ\text{C})$$

$L = -.0065; \text{K/m}$

This would put our modeled temperature at test launch at (288.08K) **14.93°C**

During comp the temperature at ground level was around **34°C**

A variation of around 20K or so.

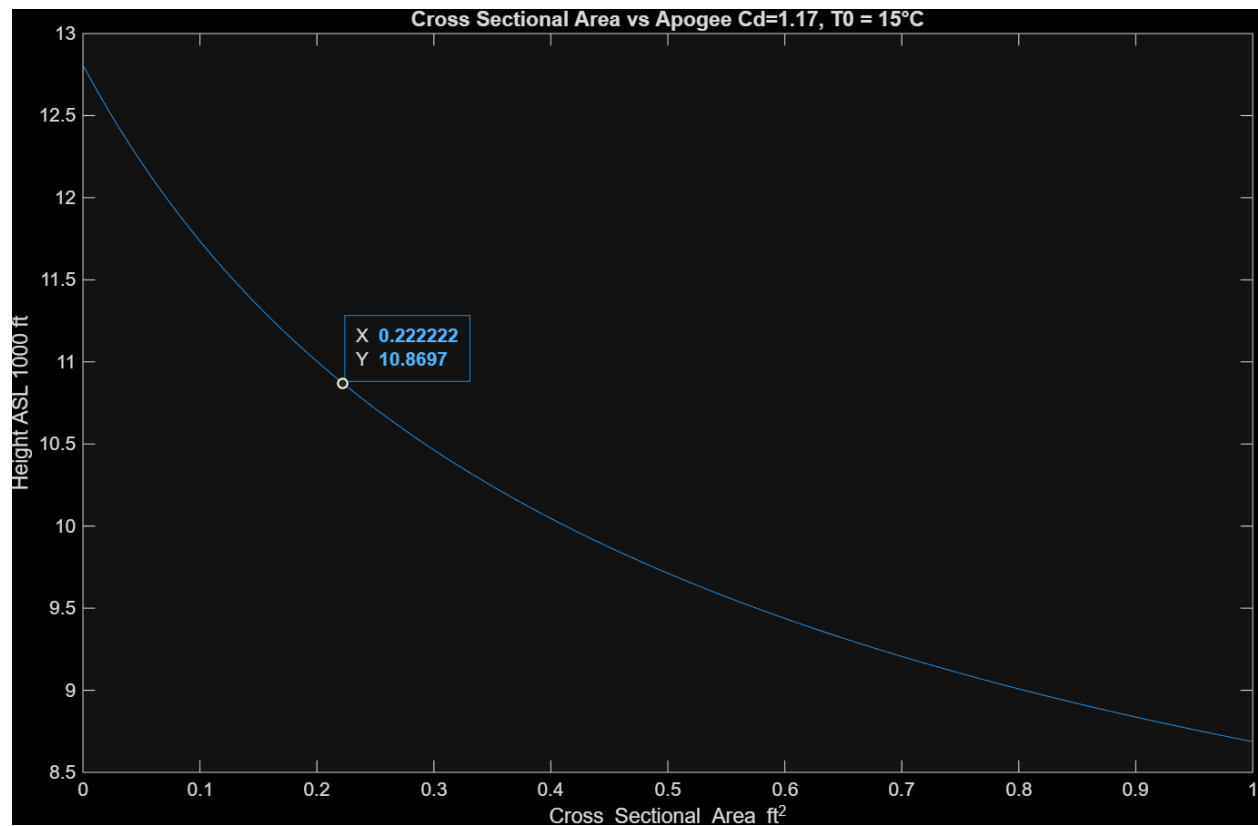


Pretty linear variation: 7 ft/K

Which means our final altitude could potentially be 140 ft higher than expected.

## Section 1: Further Determination of Drag Coefficient

We're going with 1.17 now.



**Cross Sectional Area: .222 ft<sup>2</sup>**

Maximum Force: **618.78 N**

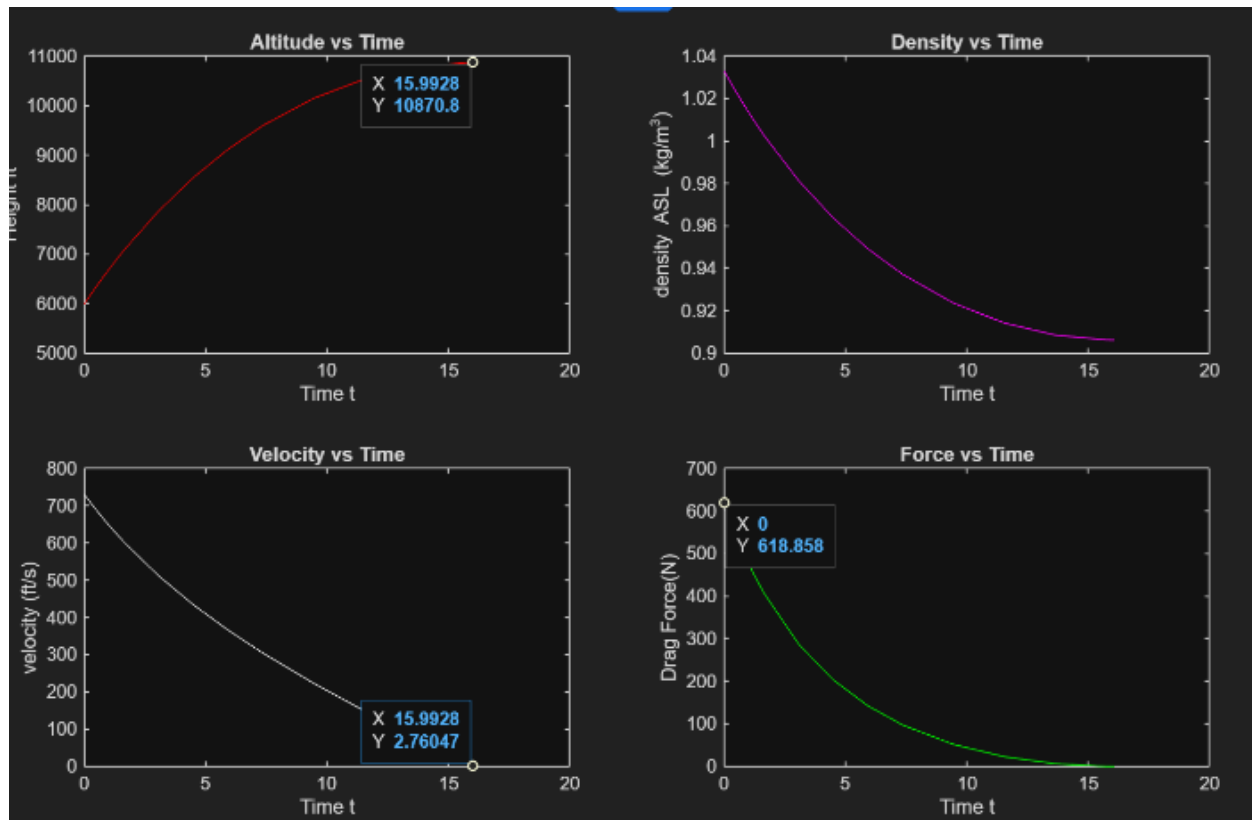
**Week of 9/29/2025**

Preliminary Analysis of Airbrakes Complete.

CD= 1.3 -> **1.17**

Surface Area (ft<sup>2</sup>): .202 -> **.222**

Maximum Force (ft<sup>2</sup>): 625.73 N -> **618.85N**



**WEEK of ?????**

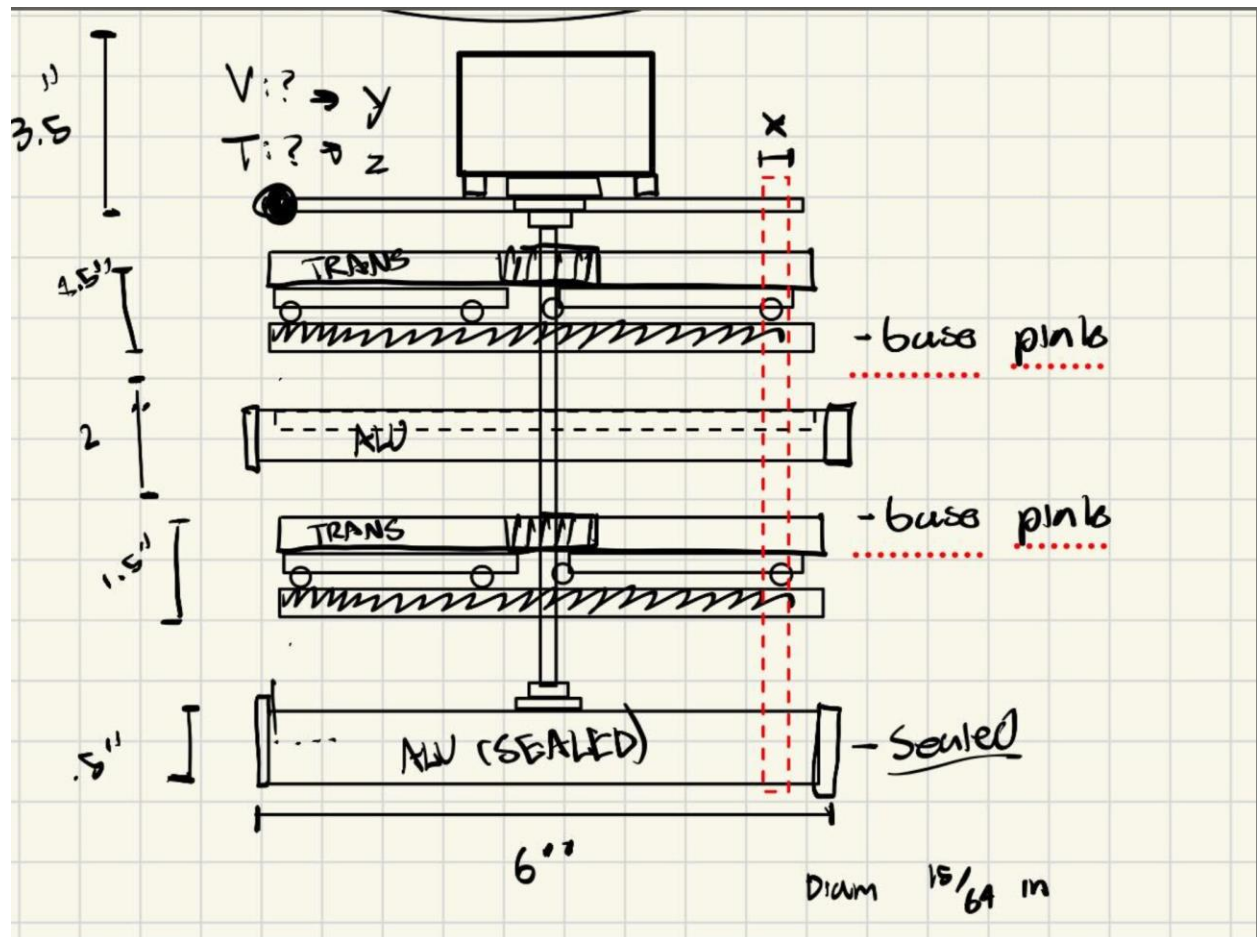
**Three Integration Bulkheads:**

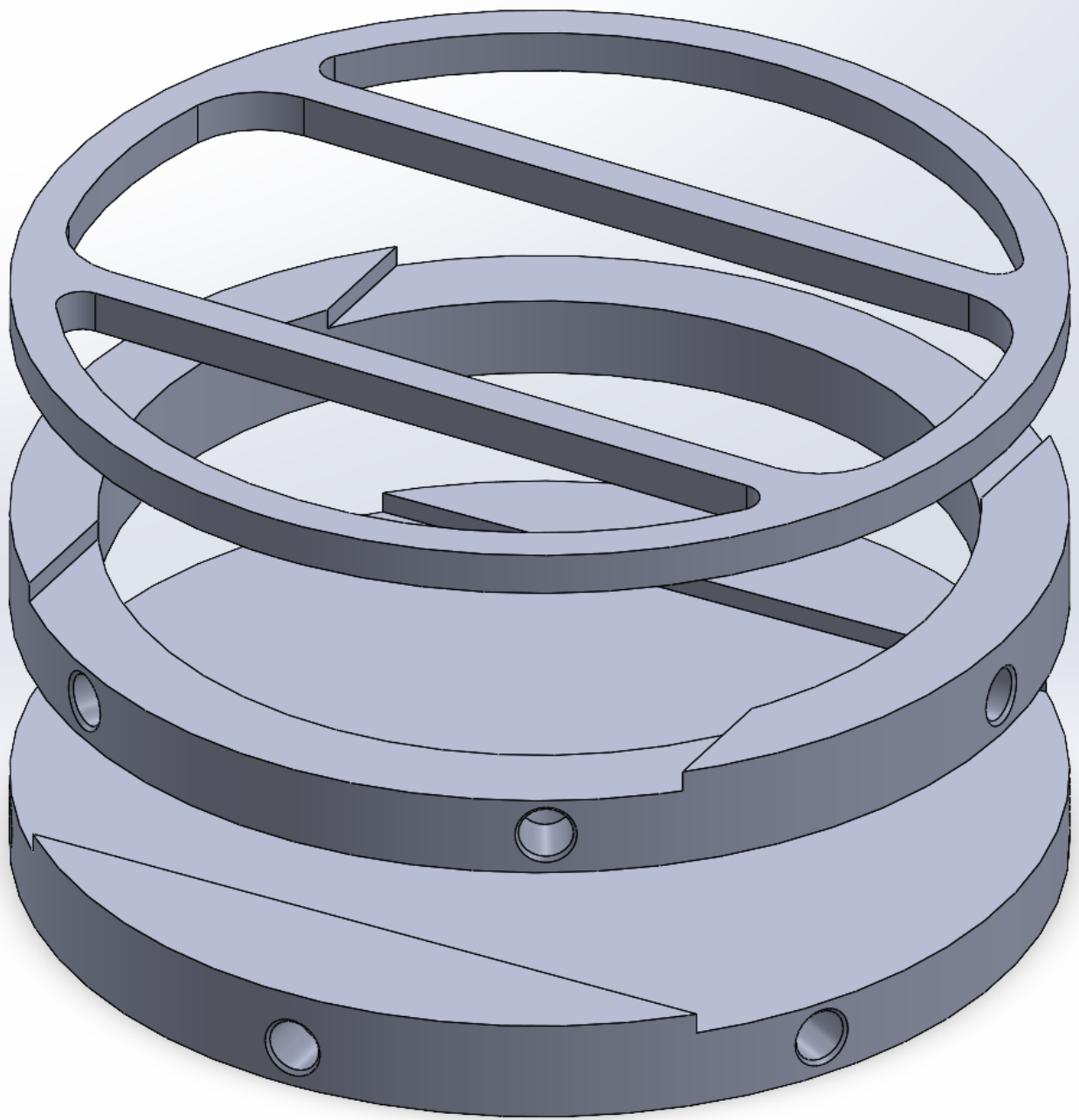
**Bottom Sealed Bulkhead**

**Middle Bulkhead**

**Motor Retaining**







*Sketch of integration mechanism layers*

**WEEK OF 10/6/2025**

Determining minimum fastener requirement.

Rings and leafs will be machined from 7075 aluminum.

Preliminary Mass estimate: **4.33 lb**

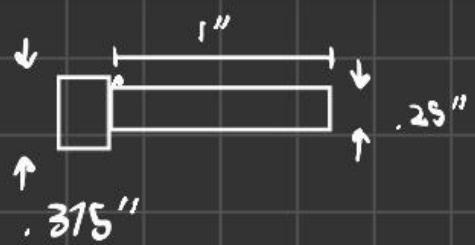
$$\begin{aligned}
 m_{sys} &= \underbrace{0.102(2\pi(3^2 - 2.5^2)(1))}_{1.7624} + \underbrace{\pi(3^2 - 2.75^2)(.25)}_{.115} + \underbrace{9\pi(.25)}_{.72} + \underbrace{(1)}_{1} + \underbrace{(28.8)(.25)(0.102)}_{.7399} \\
 &= (4.33 \text{ lbf})
 \end{aligned}$$


At 20g that is **86.6 lbf** of shear.

During maximum deployment the brakes apply 620N or 139.3 lbf of force. In addition to the weight of the system itself that corresponds to **143.66 lbf** of shear force.

Black 1/4"-20 screws: <https://www.mcmaster.com/91251A542/>

Black Oxide Alloy Steel.

$$\begin{aligned}
 \sigma_u &= 170 \text{ ksi} \\
 \tau_u &= (.6) \sigma_u = 102 \text{ ksi} \\
 A &= \pi \left( \frac{.25}{2} \right)^2 = \frac{\pi}{64} \approx 0.05 \text{ in}^2
 \end{aligned}$$


$$\begin{aligned}
 V_{max} &= \tau_u A \\
 V_{max} &= \underline{5007 \text{ lbf}}
 \end{aligned}$$

The lower bound on number of screws is set by limiting the DOF of the airbrakes in the LV, as opposed to shear strength.

One screw can support the maximum loading experienced by the system with a FoS of 34.7.

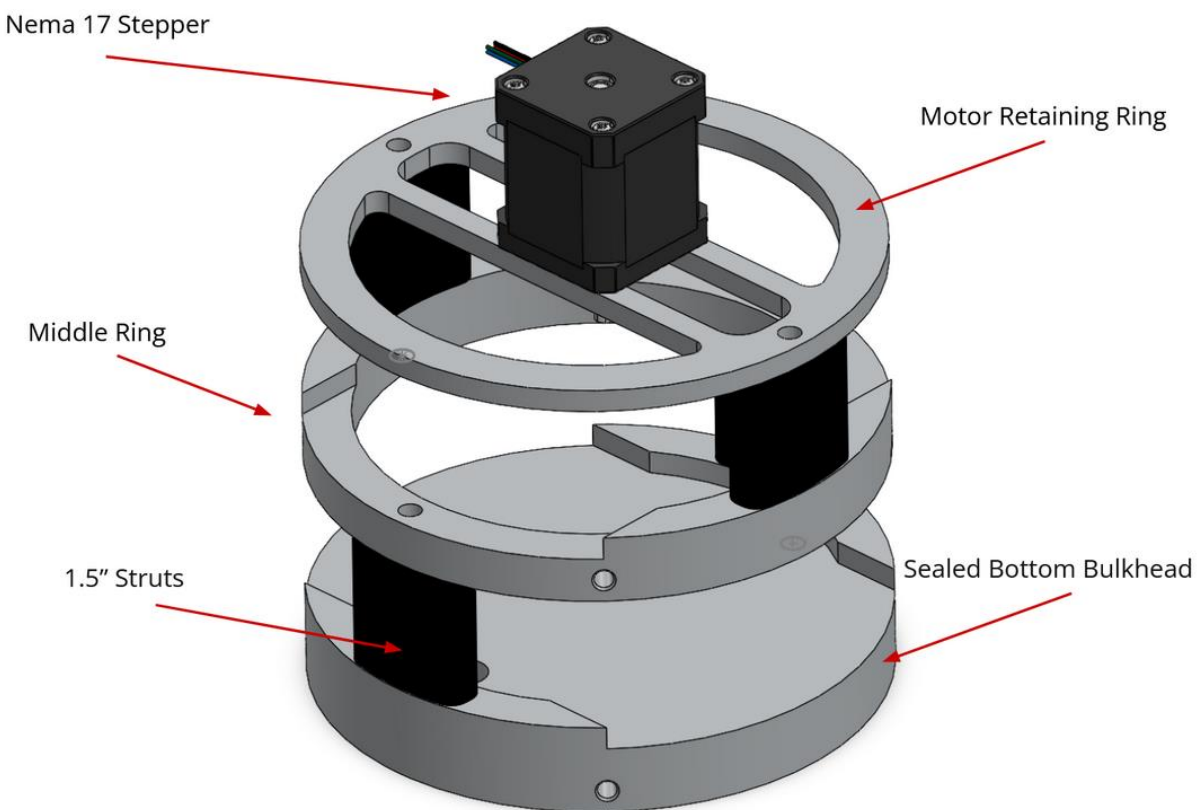
4 screws 2x ring will be sufficient.

## WEEK 10/13/2025/2025

The Integration for the system must meet the following requirements:

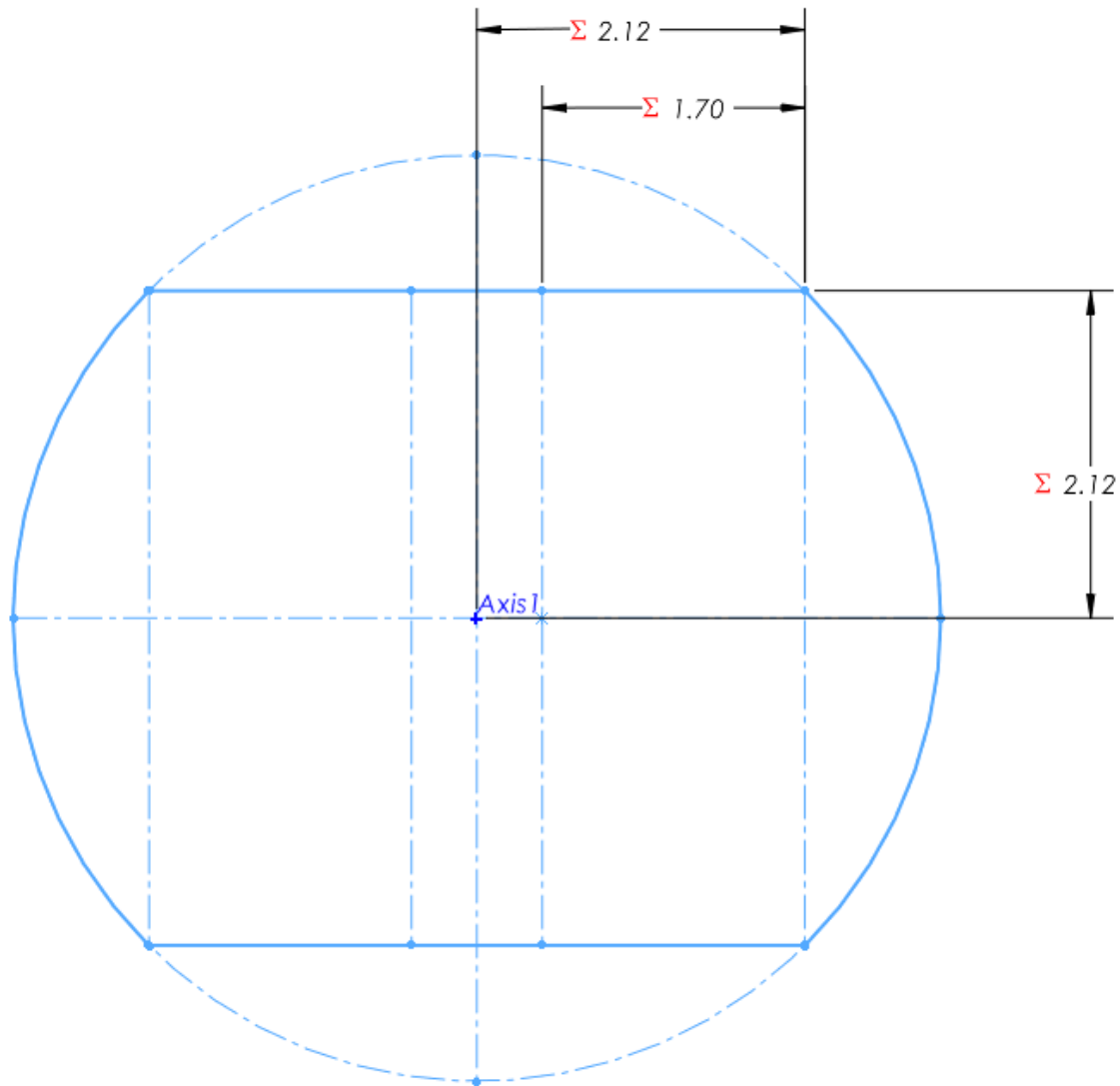
Must be able to support the maximum aerodynamic load from the breaks and transfer that load to the Air frame. The maximum predicted load on the system is 140 lbs of force, 70 lbs for each system.

Must be able to react against the torque exerted by the of the motor: 1.6 lb in.



System Mass: **2.3 lbs**

System Height: **6.89"**



There are no "free area" given the current geometry of the breaks. This is the area-maximized for the breaks. We could consider slightly reducing the width of the plates. While this would reduce the braking surface area, it would make a direct internal pass through far more feasible. Barring a reduction in break width, an internal pass through would required routing past the second set of brakes.

**WEEK 10/20/2025/2025**

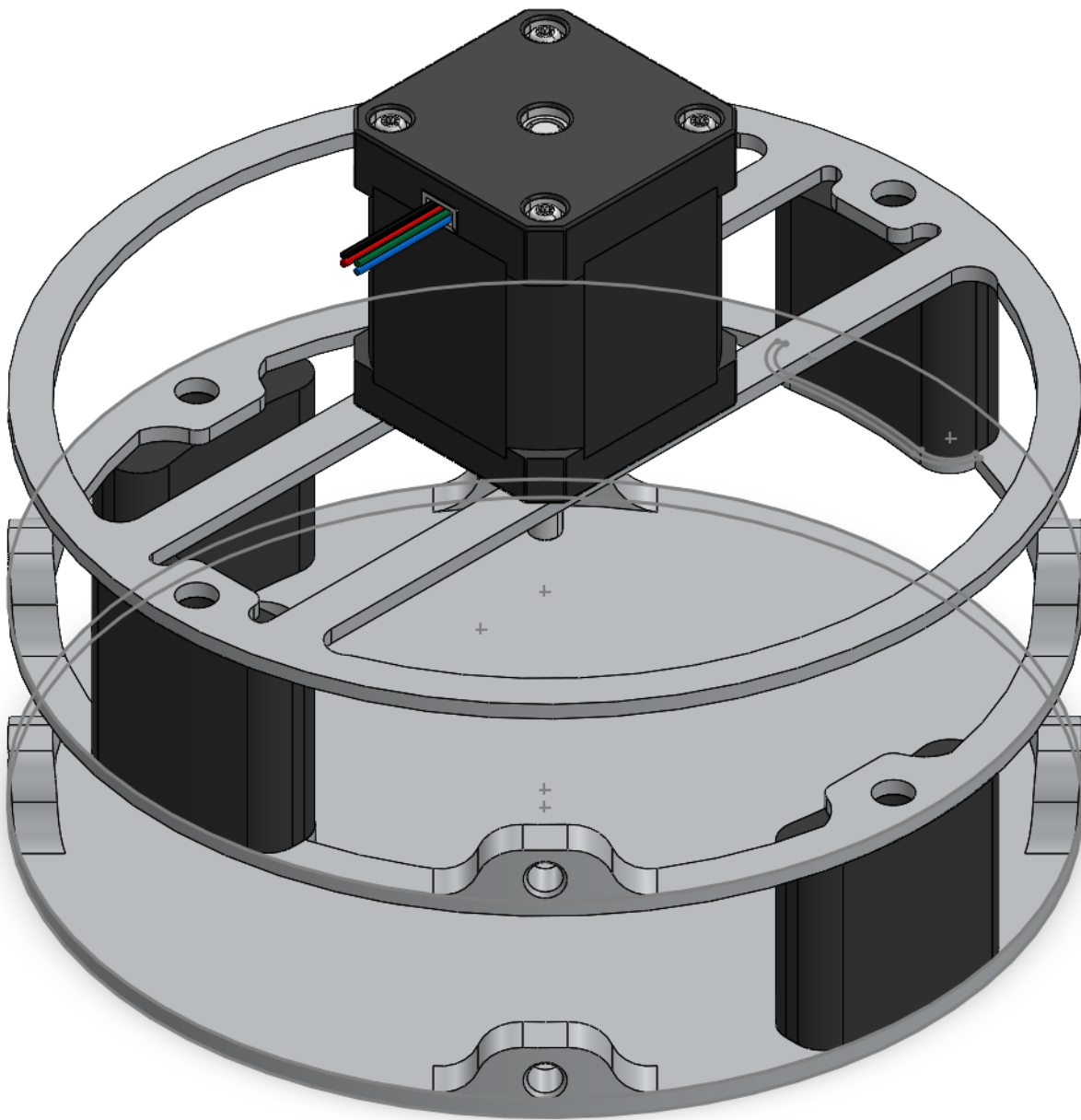
Implemented first round of IDR fixes. Mass reductions on all bulkheads. Decrease in strut height to 1.25". Final Height will be determined during integration meetings with the airbrakes mechanism.

Decreased bulkhead thickness and increased thickness at fastener location.

The Fasteners may need to have smaller securing sections depending on the Airbrake thickness.

System Mass: 2.3 → **0.7 lbs**

System Height: 6.89" → **4.71"**



## **WEEK 10/19/2025/2025**

This week's focus is integration of the electrical pass through using a top and bottom Amphenol Connector. The requirements are as follows:

- Must be able to integrate with the marotta valve
- Must be able to maintain a hermetic seal
- Must be able to support a pin count of at least (10?)
- Meet size constraints

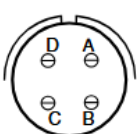
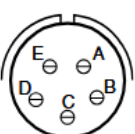
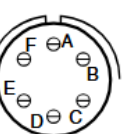
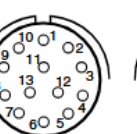
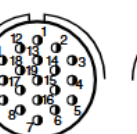
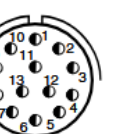
The Marotta valves use connectors from the DS38999 Connector.

It would be beneficial to be able to use a feed through connector that has port mates on both ends of the connector.

The [BLiMS electrical pass-through](#) considered a BTV 07 Jam Nut Receptacle Amphenol Feed through Connectors.

They utilized a Hermetic 07 Jam Nut Receptacle. Military (D38999/23), Commercial (TVS07)

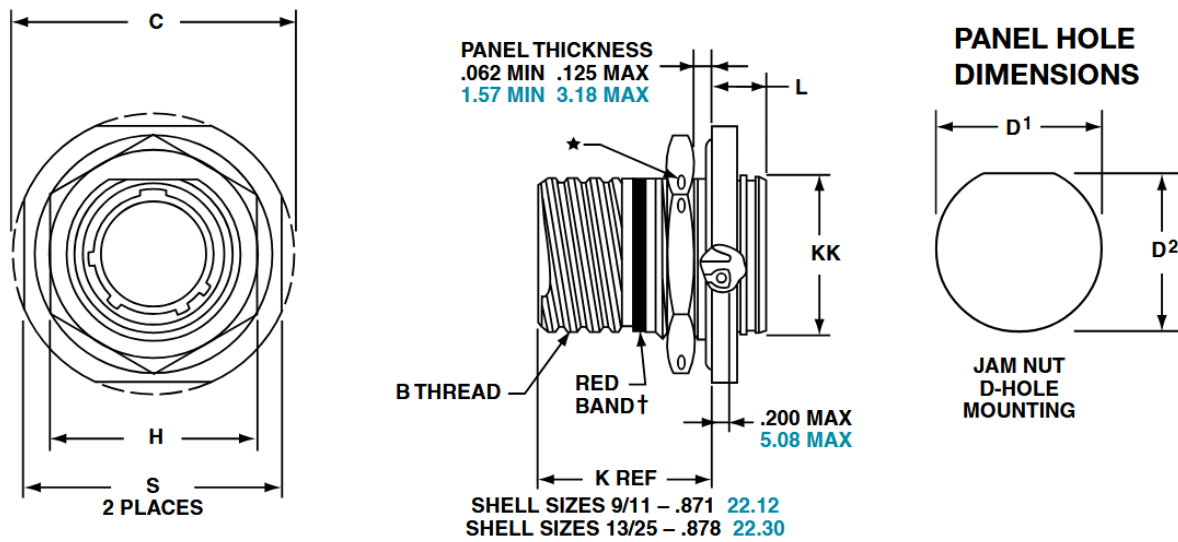
As for the insert arrangement they used 11-35

| Shell Size & Insert Arrg. for: |  |  |  |  |  |  |
|--------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Series II JT</b>            | 10-4                                                                              | 10-5                                                                              |                                                                                   | 10-13                                                                              |                                                                                     | 10-35                                                                               |
| <b>Series I LJT</b>            | 11-4                                                                              | 11-5                                                                              | 11-6                                                                              | 11-13                                                                              |                                                                                     | 11-35                                                                               |
| <b>Series III TV</b>           | 11-4                                                                              | 11-5                                                                              |                                                                                   |                                                                                    | 11-19 HD                                                                            | 11-35                                                                               |
| Service Rating                 | I                                                                                 | I                                                                                 | I                                                                                 | M                                                                                  | N                                                                                   | M                                                                                   |
| Number of Contacts             | 4                                                                                 | 5                                                                                 | 6                                                                                 | 13                                                                                 | 19                                                                                  | 13                                                                                  |
| Contact Size                   | 20                                                                                | 20                                                                                | 20                                                                                | 22M                                                                                | 23                                                                                  | 22D                                                                                 |

13 Contacts, 22D contact size

Here are the shell size options for the TV07/ Note the allowable Panel Thickness is .625 - .125". (Bulkhead thickness 0.1")





† Red band indicates fully mated

★ .059 dia min.

1.5 dia min. 3 lockwire holes Formed lockwire hole design (6 holes) is optional.

Inches

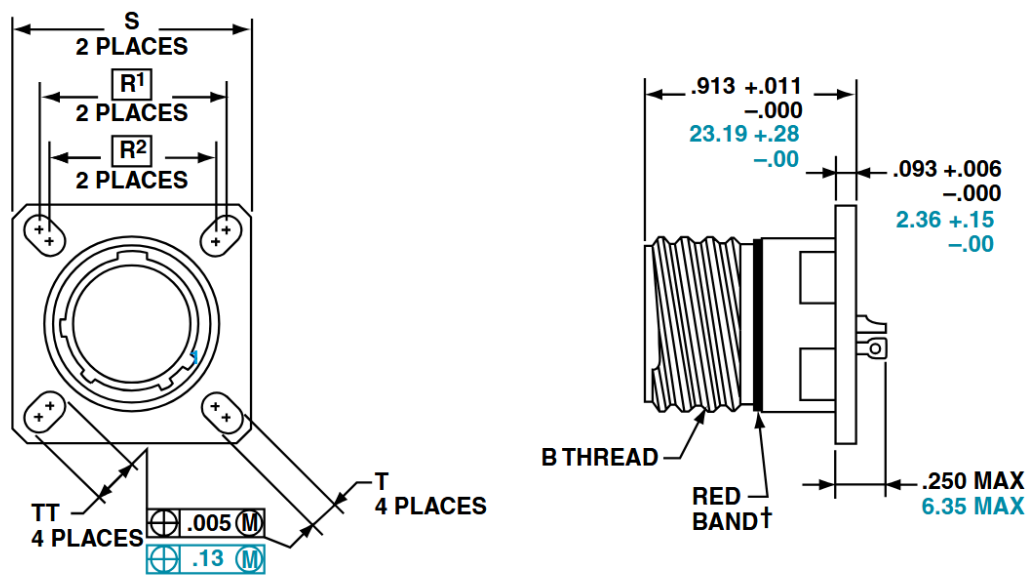
| Shell Size | MS Shell Size code | B Thread Class 2A 0.1P-0.3L-TS (Plated) | C Max | D <sup>1</sup> +.010<br>–.000 | D <sup>2</sup> +.000<br>–.010 | H Hex +.017<br>–.016 | L Max | S ±.010 | KK +.011<br>–.000 |
|------------|--------------------|-----------------------------------------|-------|-------------------------------|-------------------------------|----------------------|-------|---------|-------------------|
| 9          | A                  | .6250                                   | 1.199 | .693                          | .657                          | .875                 | .357  | 1.062   | .642              |
| 11         | B                  | .7500                                   | 1.386 | .825                          | .770                          | 1.000                | .357  | 1.250   | .766              |
| 13         | C                  | .8750                                   | 1.511 | 1.010                         | .955                          | 1.188                | .357  | 1.375   | .892              |
| 15         | D                  | 1.0000                                  | 1.636 | 1.135                         | 1.085                         | 1.312                | .357  | 1.500   | 1.018             |
| 17         | E                  | 1.1875                                  | 1.761 | 1.260                         | 1.210                         | 1.438                | .357  | 1.625   | 1.142             |
| 19         | F                  | 1.2500                                  | 1.949 | 1.385                         | 1.335                         | 1.562                | .381  | 1.812   | 1.268             |
| 21         | G                  | 1.3750                                  | 2.073 | 1.510                         | 1.460                         | 1.688                | .381  | 1.938   | 1.392             |
| 23         | H                  | 1.5000                                  | 2.199 | 1.635                         | 1.585                         | 1.812                | .381  | 2.062   | 1.518             |
| 25         | J                  | 1.6250                                  | 2.323 | 1.760                         | 1.710                         | 2.000                | .381  | 2.188   | 1.642             |

Smallest Shell Size (9) may still present substantial integration difficulties

Additional Options:

Hermetic Box Mounting Receptacle

Military (D38999/21), Commercial (TVPS02)



† Red band indicates fully mated

NOTE: Consult Amphenol Aerospace for availability of non-glass-sealed versions with printed circuit tail contacts.

Inches

| Shell Size | MS Shell Size Coded | B Thread 0.1P-0.3L-TS (Plated) | R1    | R2    | S $\pm .010$ | T $\pm .008$ | TT $\pm .008$ |
|------------|---------------------|--------------------------------|-------|-------|--------------|--------------|---------------|
| 9          | A                   | .6250                          | .719  | .594  | .938         | .128         | .216          |
| 11         | B                   | .7500                          | .812  | .719  | 1.031        | .128         | .194          |
| 13         | C                   | .8750                          | .906  | .812  | 1.125        | .128         | .194          |
| 15         | D                   | 1.0000                         | .969  | .906  | 1.219        | .128         | .173          |
| 17         | E                   | 1.1875                         | 1.062 | .969  | 1.312        | .128         | .194          |
| 19         | F                   | 1.2500                         | 1.156 | 1.062 | 1.438        | .128         | .194          |
| 21         | G                   | 1.3750                         | 1.250 | 1.156 | 1.562        | .128         | .194          |
| 23         | H                   | 1.5000                         | 1.375 | 1.250 | 1.688        | .154         | .242          |
| 25         | J                   | 1.6250                         | 1.500 | 1.375 | 1.812        | .154         | .242          |

Week of 11/1/25

Lintech Mini Rail (MR) series Integration: Lintech MR 7 ML

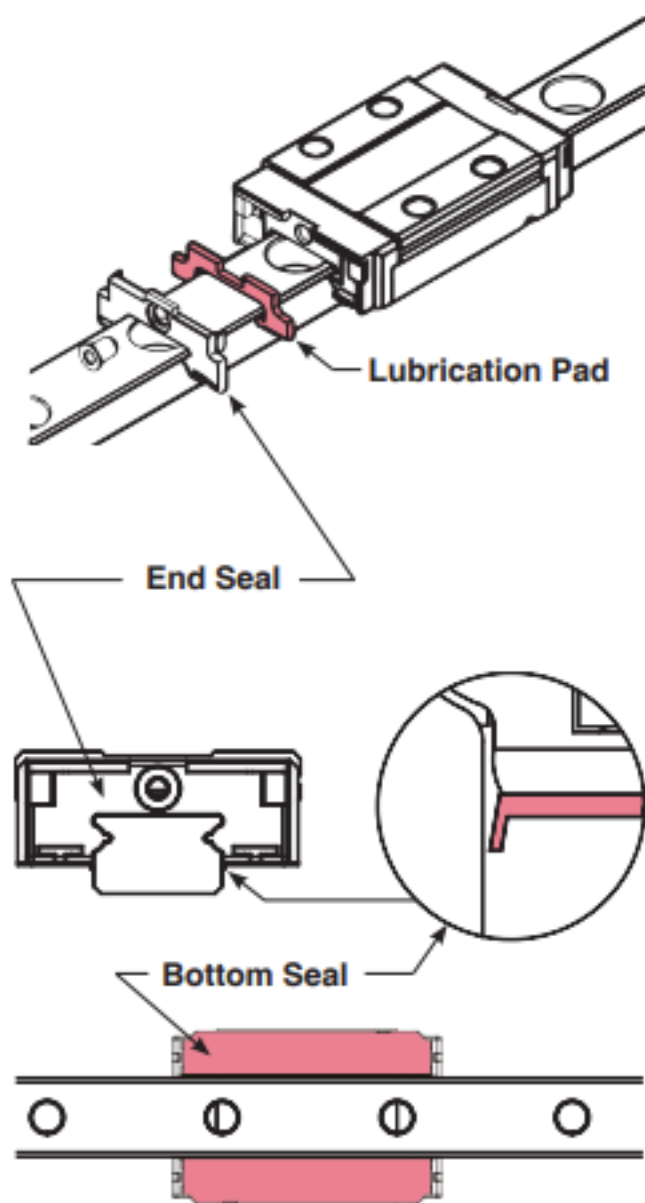
[MR series Miniature Linear Guide | Lintech](#)

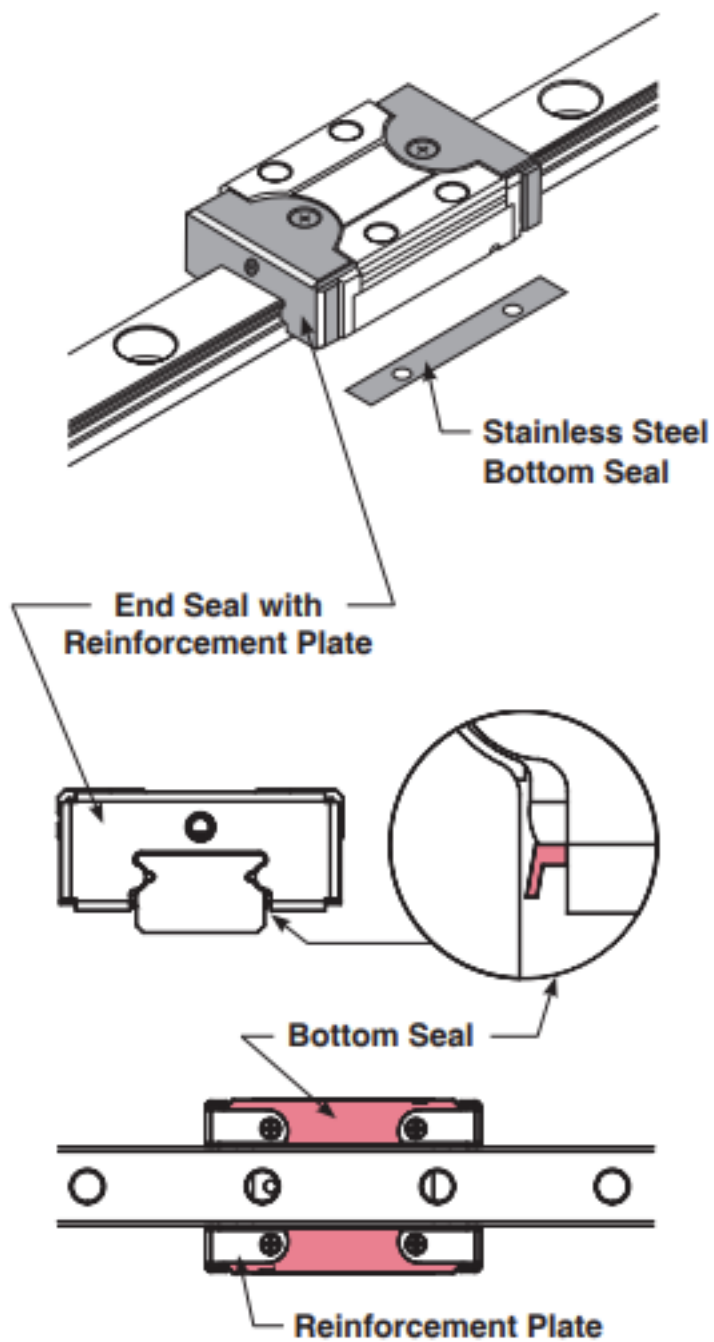
The profile-rail system consists of a linear block with two rows of recirculating balls. Aimed for small-space applications requiring light to medium load capacities while maintaining precise position accuracy which makes them ideal for the airbrake system.

Based on Sealing system the max speed and acceleration of the system is changed. Based on the Airbrake Mechanism Calculations the constant linear acceleration of the leaf is  $17.272 \text{ m/s}^2$  and  $0.8636 \text{ m/s}$

Long-term motion is not a primary consideration for the airbrake system. Competition environment makes dust resistance a factor.

Based on the system kinematics and environment I believe ZU - End and Bottom Seals with Lubrication Pads is the optimal lubricant solution for the rails. (ZU/ZUE).



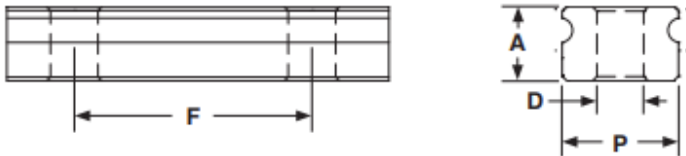


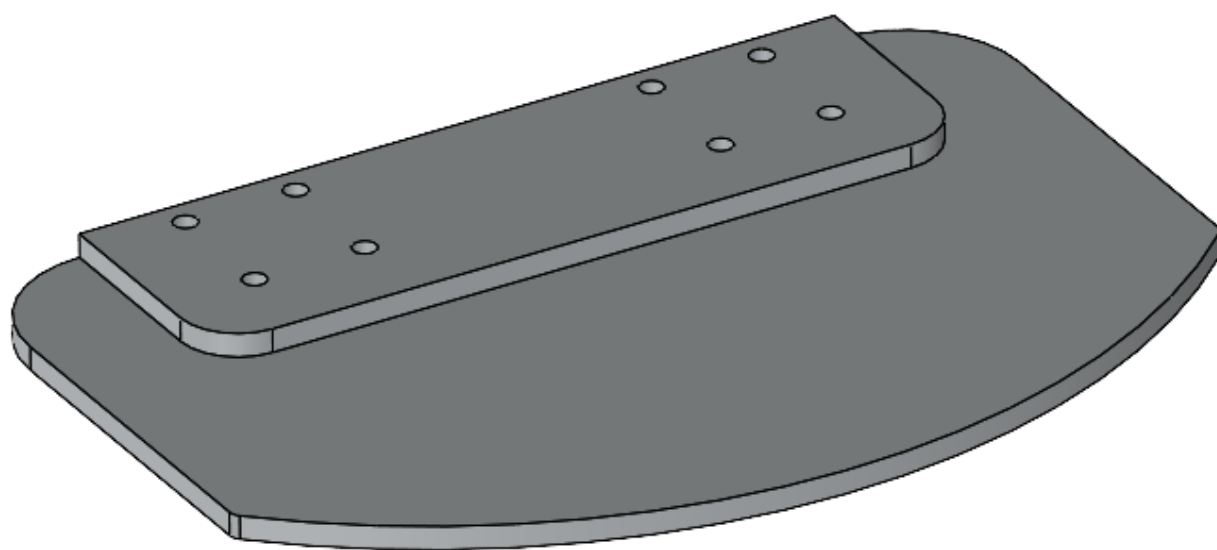
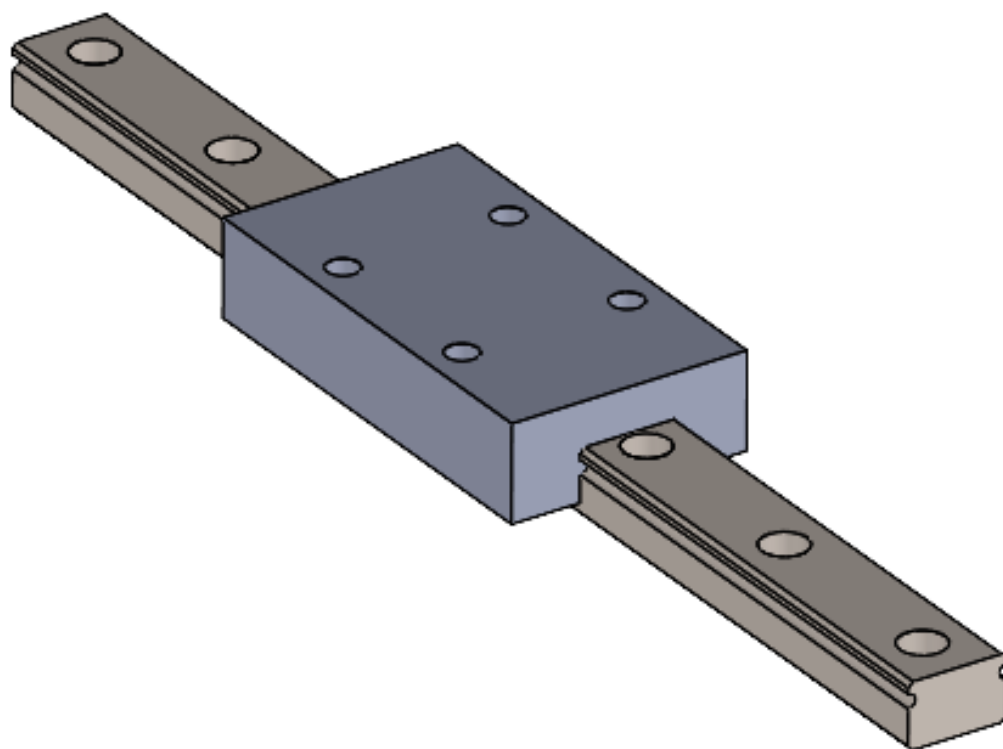
## Speed & Acceleration - MR

| Definition                                    | End and Bottom Seal Version |       |       |       |         |
|-----------------------------------------------|-----------------------------|-------|-------|-------|---------|
|                                               | SS/ZZ                       | SU/ZU | EE/EZ | EU/UZ | SUE/ZUE |
| <b>Max Speed</b> (m/sec)                      | 3                           | 3     | 5     | 5     | 5       |
| <b>Max Acceleration</b> (m/sec <sup>2</sup> ) | 250                         | 250   | 300   | 300   | 300     |

Rail: Standard Rail Tapped from Bottom - MRU.

## Standard Rail Tapped from Bottom - MRU

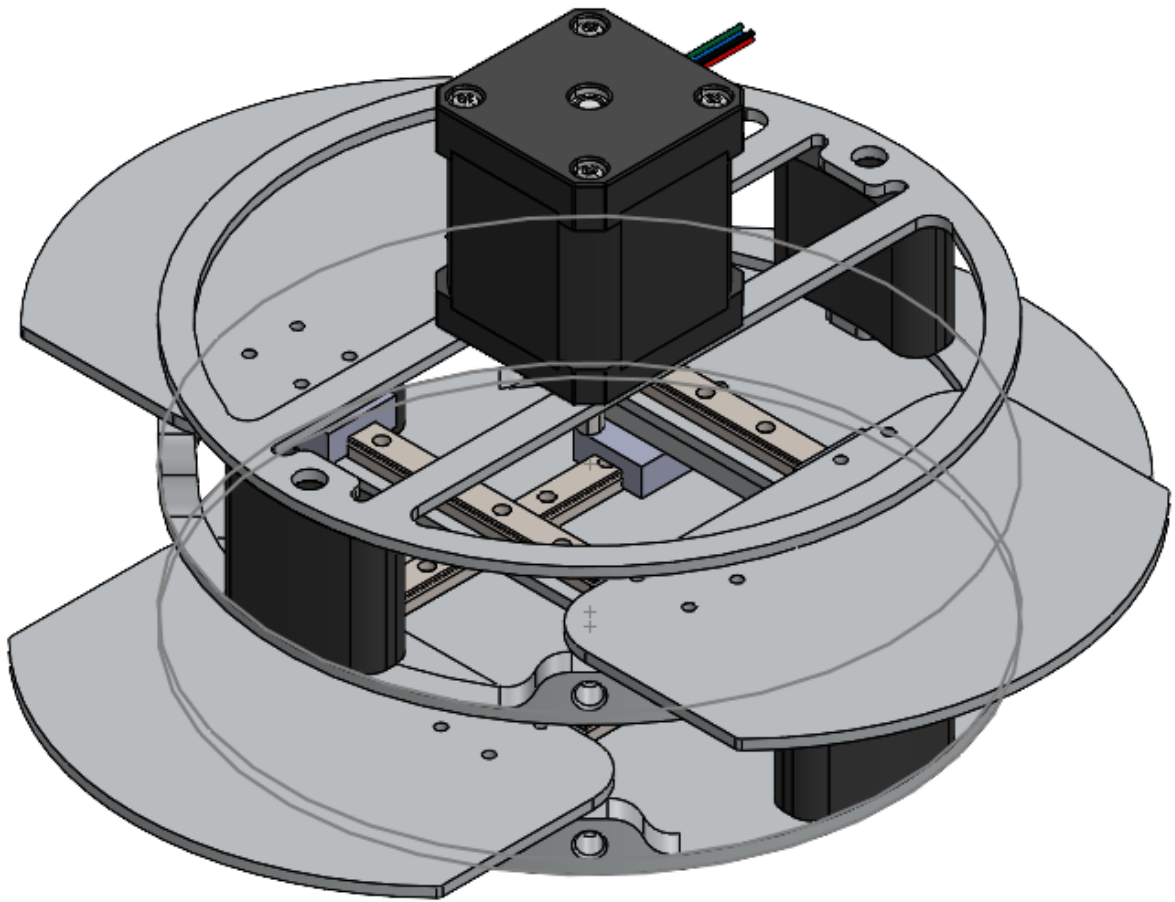
| Rail Size | (mm)        |    |     |    |                                                                                                                          |
|-----------|-------------|----|-----|----|--------------------------------------------------------------------------------------------------------------------------|
|           | D           | P  | A   | F  |                                                                                                                          |
| MRU 3M    | M1.6 x 0.35 | 3  | 2.6 | 10 | <p><b>Rail Tapped from Bottom</b></p>  |
| MRU 5M    | M3 x 0.5    | 5  | 3.5 | 15 |                                                                                                                          |
| MRU 7M    | M3 x 0.5    | 7  | 4.7 | 15 |                                                                                                                          |
| MRU 9M    | M4 x 0.7    | 9  | 5.5 | 20 |                                                                                                                          |
| MRU 12M   | M4 x 0.7    | 12 | 7.5 | 25 |                                                                                                                          |
| MRU 15M   | M4 x 0.7    | 15 | 9.5 | 40 |                                                                                                                          |

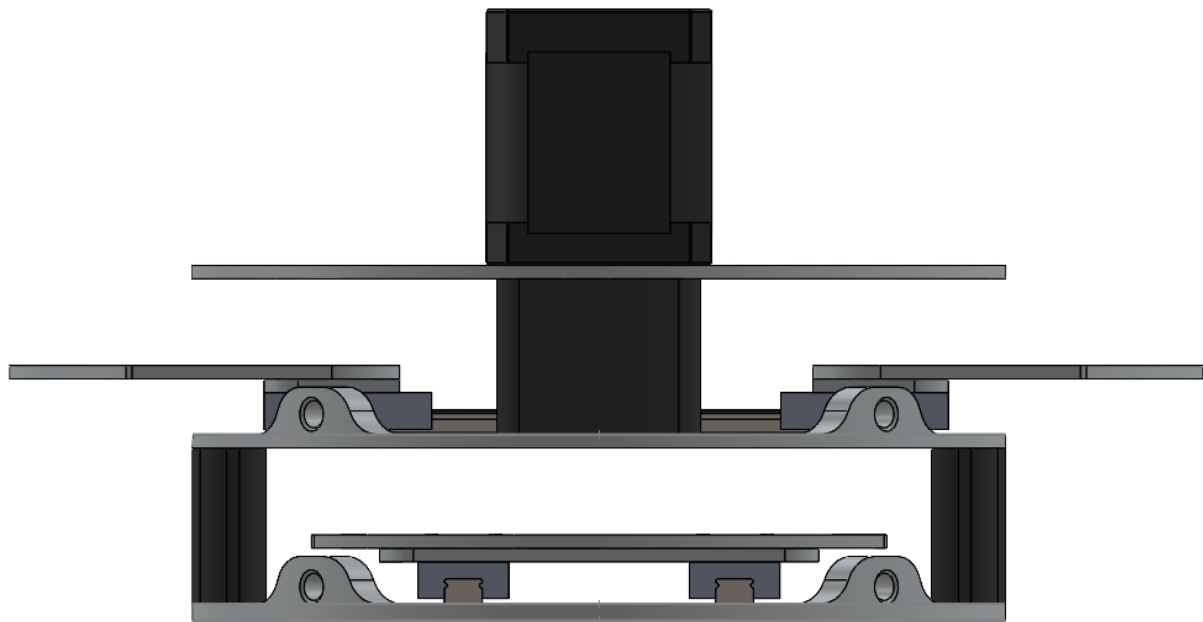


MRU 7M Rail and MR 7 ML Linear Block  
Mounting Offset

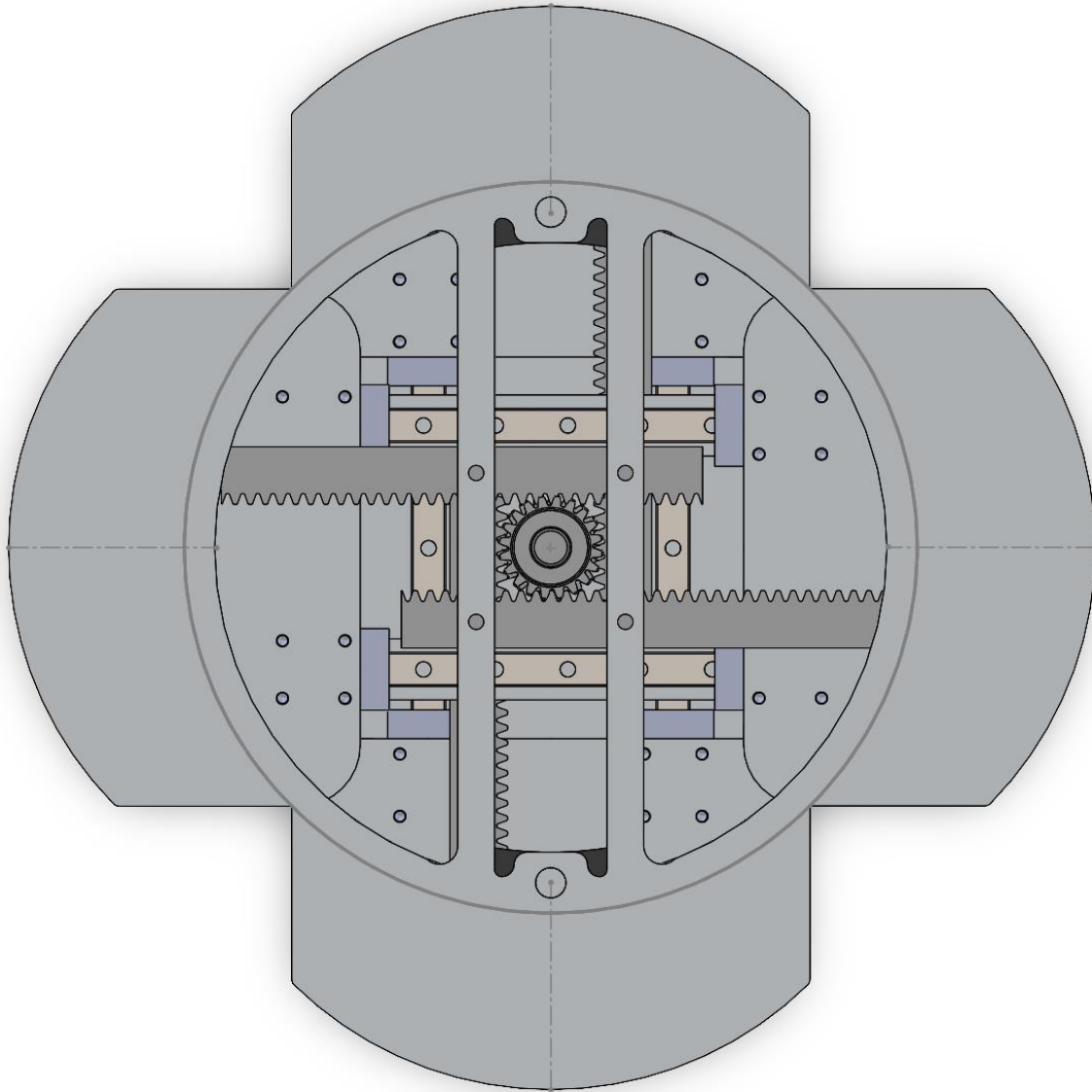
Break Leaf w/







Transmission Concept:



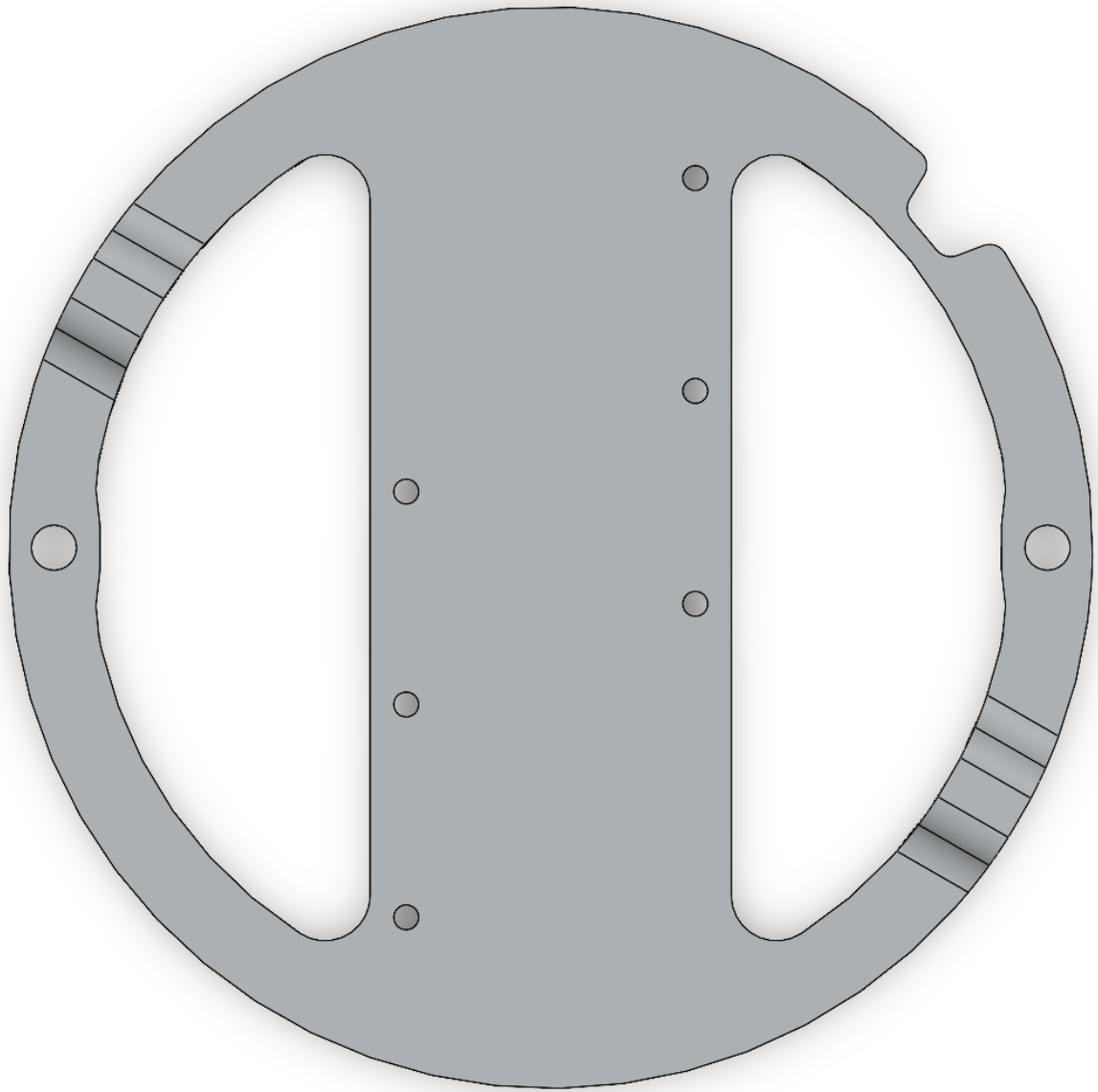
[20251104-2317-24.6815340.mp4](#)

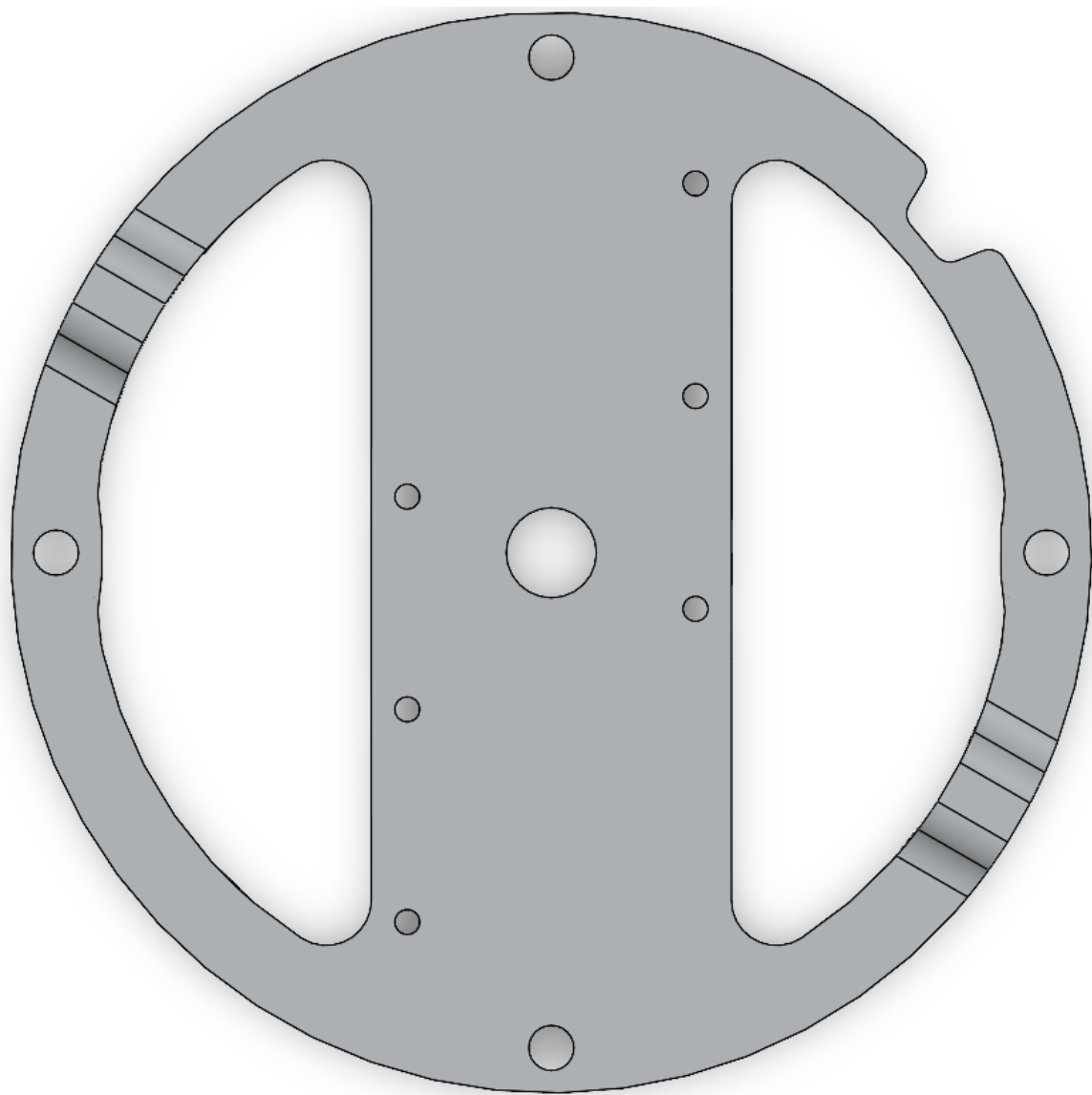


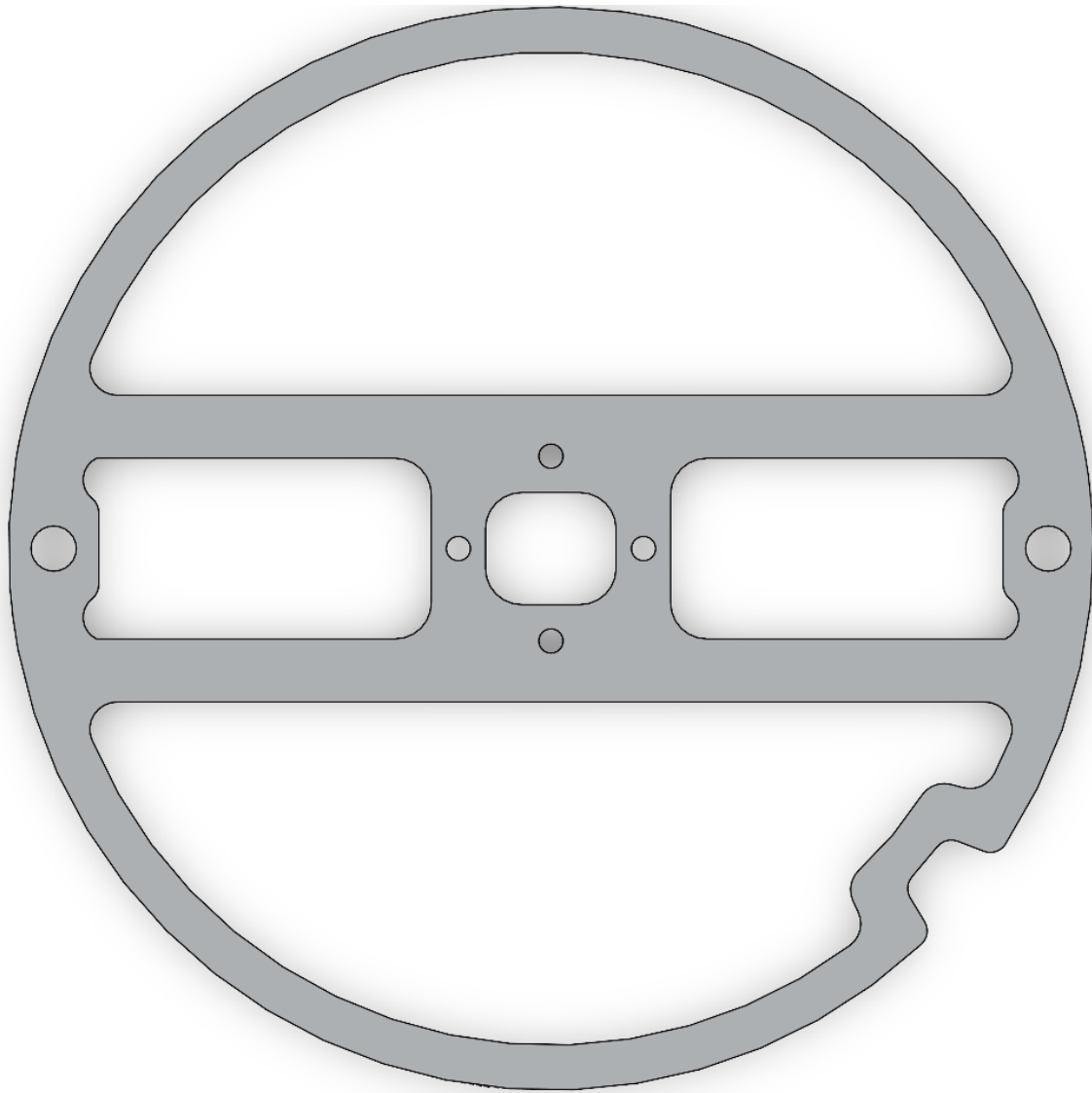
Week of 11/9/2025 - Week of FDR:

We will no longer be implementing amphenol feed-through bulkheads. This is because there is no longer a need to seal the bottom bulkhead. The air brake system should not fail if exposed to PRV, and the motor controller should not be damaged by the gas either. In case of PRV the airbrakes are no longer useful anyway. AV Bay will now seal against PRV. We do not need amphenols on the airbrakes and instead will just be using notches for the wire to pass through.

Also the leaves will now only slide on one rail, and will have one strut in the middle of each bulkhead. Also instead of having 4 holes on each bulkhead we'll only use 4 in total, with two pairs of holes clocked 90 deg relative to one another. The clocking of the holes relative to the bulkheads themselves is primarily driven by the need to separate the fasteners from the air frame apertures as much as possible.







Also [heat set inserts](#) and socket head [1/4"-20](#) have been selected.

Wiring route:

Upper Bulkhead Notch

I

Strut nub

|

Middle Notch

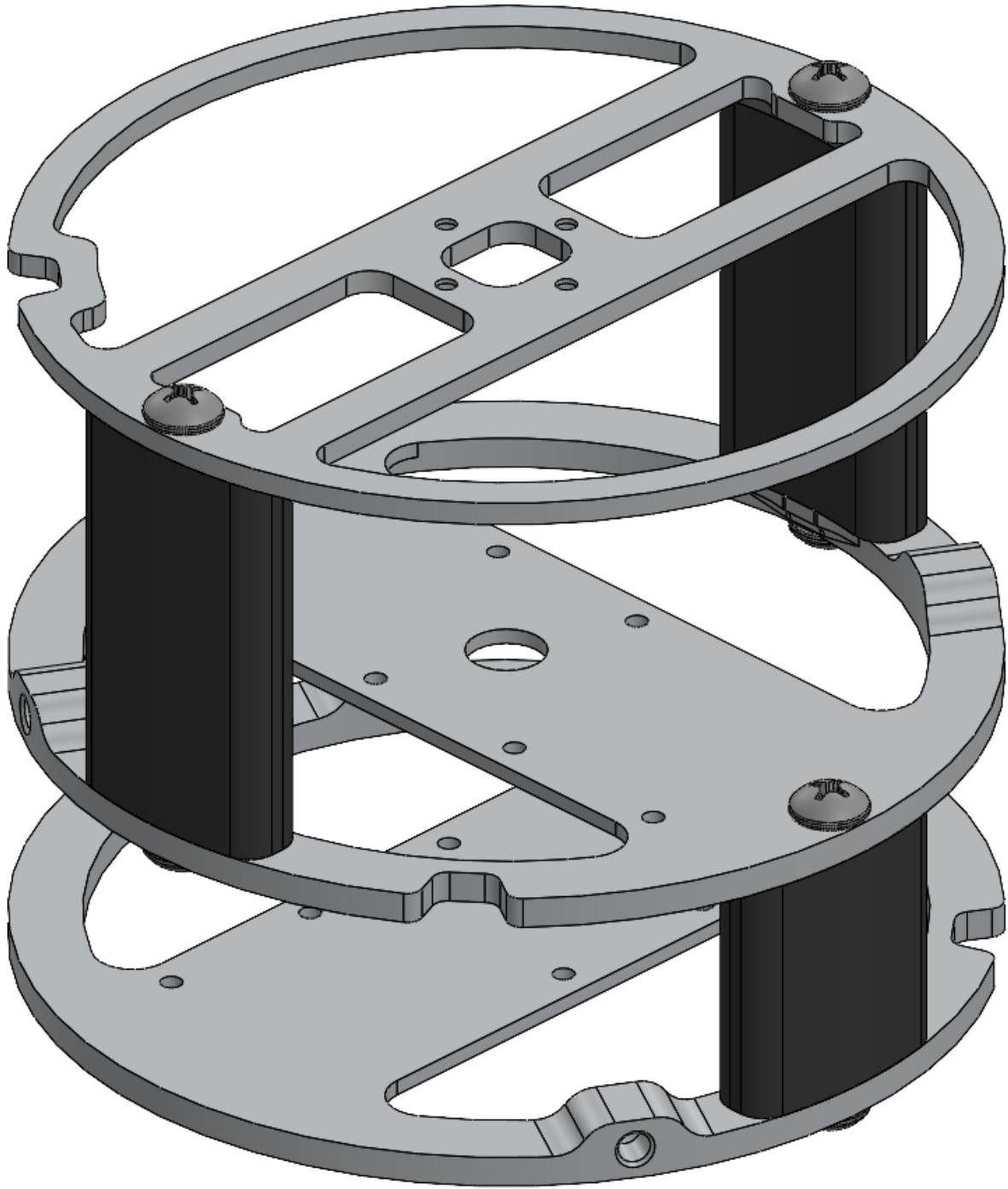
|

Strut Nub

|

Bottom Notch.

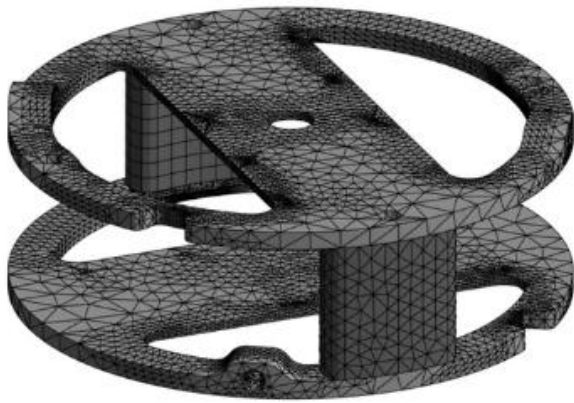




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**Integrated Structural Analysis.**



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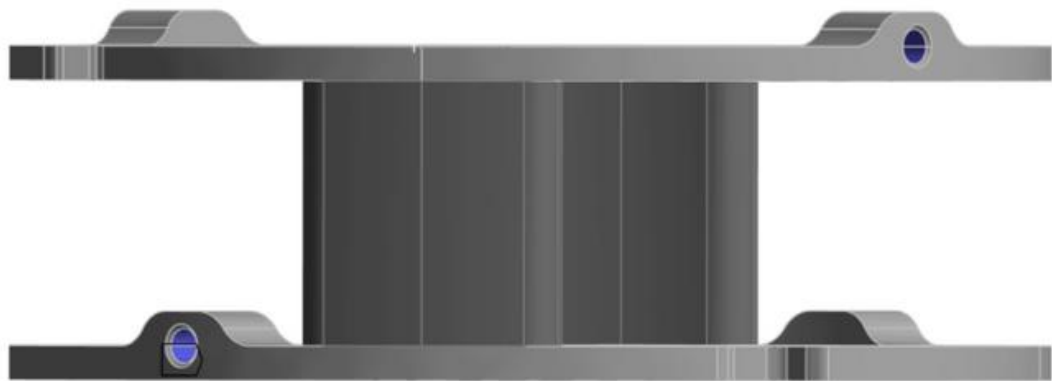
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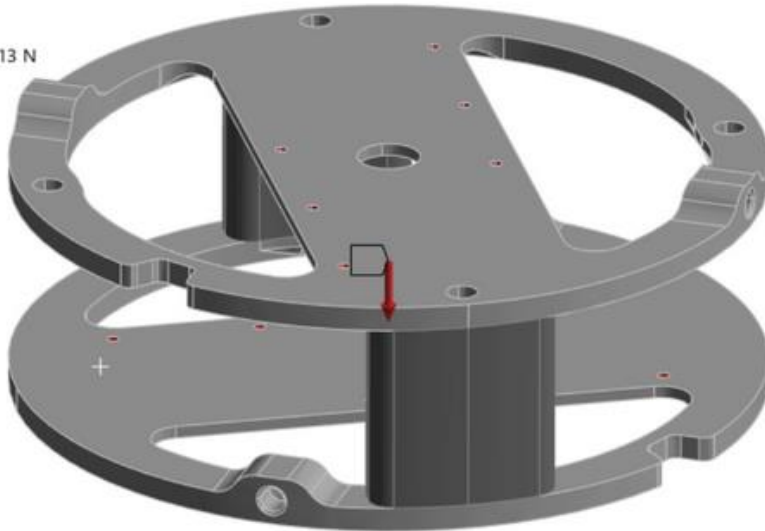
Fixed Support



Remote Force: 1550. N  
 Components: 1.2484e-012, -1550., -5.8782e-013 N  
 Location: 0.705, 0.92765, 1.3388 m

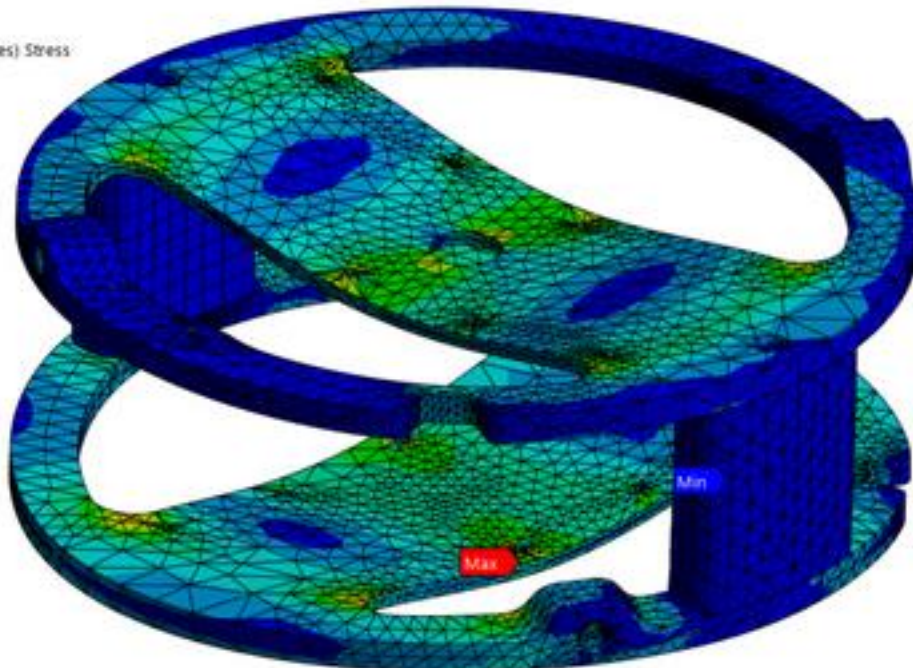
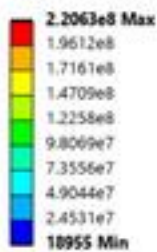


Adding Force where  
 rails transmits load  
 to bulkheads



#### B: Integration Structure

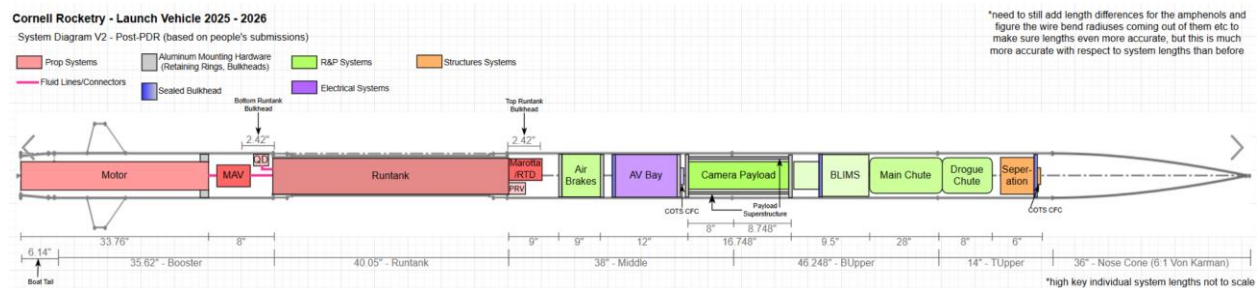
Equivalent Stress  
 Type: Equivalent (von-Mises) Stress  
 Unit: Pa  
 Time: 1 s  
 11/15/2025 2:05:21 AM



Airbrakes by design effect a significant aerodynamic pressure on the surface of the launch vehicle. The obvious effect is that the center of pressure is shifted in the direction of the

airbrakes position during deployment. In order to meet stability requirements we must quantify exactly how airbrakes effect the center of pressure of the vehicle.

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STRUCT<br>15 | Launch vehicles shall remain "stable" for the entire ascent. Stable is defined as maintaining a static margin of at least 1.5 to 2 body calibers, regardless of CG movement due to depleting consumables and shifting center of pressure (CP) location due to wave drag effects (which may become significant as low as 0.5 M). Not falling below 2 body calibers will be considered nominal, while falling below 1.5 body calibers will be considered a loss of stability. |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



The airbrakes are nominally located 85.12" from the base of the LV.

CASE STUDY 1: Space Cowboys (Mississippi State): <https://arc.aiaa.org/doi/epdf/10.2514/6.2024-85628>

The air brake system prototype was chosen to be placed behind the center of gravity on the flight vehicle to minimize instability during the rocket's flight.

System was NOT flown.

L2 Rocket:

84" Length

OD: 4.12"

Weight: 11.3 lbs

Top Speed: 566 ft/s

Drag Output (Predicted: 32 lbf)

Not representative of hybrid rocket. No formal calculations for CP location shift. Just positioned the airbrakes aft of the CG, like fins.

CASE STUDY 2: Department of Mechanical and Process Engineering ETH Zürich:  
[https://www.eucass.eu/doi/EUCASS2019-0388.pdf?utm\\_source=chatgpt.com](https://www.eucass.eu/doi/EUCASS2019-0388.pdf?utm_source=chatgpt.com)

The airbrakes are located behind the center of pressure to guarantee stability. This paper uses the Barrowman Formulation of Center of Pressure. This method breaks down the surface of the rocket into different segments (nosecone, body, fins...) and calculates the normal force for those coefficient segments. As shown in equation [32] the position of the center of pressure is a weighted average of the center of pressure of each segmented CoP. The aerodynamic forces acting on the surface of the airbrakes are purely axial (with respect to the longitudinal axis of the rocket). However only normal forces effect the stability of the rocket. Paper claims axially symmetric system actuated by a single motor effects no normal forces on the rocket, for a small angle of attack. The normal force coefficient summation [31b] does not include a contribution from airbrakes.

Our system is actuated by a single motor and is designed to only enact an axial drag force on the vehicle. Therefore I believe our system matches the assumptions made in this analysis of CoP:

Second, according to Barrowman [2], the normal force coefficient  $C_N$  can be computed as the product of the angle of attack  $\alpha$  with the sum of the body  $C_{N_{\alpha,B}}$ , nosecone  $C_{N_{\alpha,N}}$ , fins  $C_{N_{\alpha,F}}$  and fins-body interference  $C_{N_{\alpha,T(B)}}$  drag coefficient derivatives, with a correction term  $C_{N_{\alpha 2}}$  defined by Galejs [13] to account for body lift, as

$$C_N(\mathbf{x}) = C_{N_{\alpha}} \alpha \quad (31a)$$

$$C_{N_{\alpha}} = C_{N_{\alpha,B}} + C_{N_{\alpha,N}} + C_{N_{\alpha,F}} + C_{N_{\alpha,T(B)}} + C_{N_{\alpha 2}}. \quad (31b)$$

As the airbrakes are symmetrical with respect to the longitudinal axis of the rocket and are actuated with a single motor, no normal drag is induced by the airbrakes.

Finally, the damping moment drag coefficient  $C_{damp}(\mathbf{x})$  is determined according to [18, 202]. To compute the normal torque  $\tau_N$  using the stability margin  $\bar{x}_{stab}$ , it is necessary to compute the position of the center of pressure  $\mathbf{r}_{CoP}$ . Given the respective centers of pressure  $\mathbf{r}_{CoP,i}$  of the different sections of the rocket (body, nosecone, ...) [2], it is computed as a weighted sum of each drag coefficient derivative  $C_{N_{\alpha,i}}$  as

$$\mathbf{r}_{CoP} = \frac{\sum \mathbf{r}_{CoP,i} C_{N_{\alpha,i}}}{\sum C_{N_{\alpha,i}}}. \quad (32)$$

CASE STUDY 3: Rocket Py

Because CRT is exploring Rocket Py as an alternative to other Rocket Sim Software it is worth examining how Rocket Py formulates stability effects on Airbrakes. In rocket by airbrakes inherit from the AeroSurface. Other AeroSurfaces include NoseCone, Tail, Fins (Abstract), Rail Buttons, etc... AeroSurfaces all include an evaluate center of pressure function that returns the CoP of the surface in local coordinates. The location of the CoP is always on the longitudinal axis of the vehicle ( $x = 0$ ,  $y = 0$ ).

For example this is the evaluate center of pressure for trapezoidal fins:

```
def evaluate_center_of_pressure(self):
    """Calculates and returns the center of pressure of the fin set in local
    coordinates. The center of pressure position is saved and stored as a
    tuple.

    Returns
    -----
    None
    """
    # Center of pressure position in local coordinates
    cpz = (self.sweep_length / 3) * (
        (self.root_chord + 2 * self.tip_chord) / (self.root_chord + self.tip_chord)
    ) + (1 / 6) * (
        self.root_chord
        + self.tip_chord
        - self.root_chord * self.tip_chord / (self.root_chord + self.tip_chord)
    )
    self.cpx = 0
    self.cpy = 0
    self.cpz = cpz
    self.cp = (self.cpx, self.cpy, self.cpz)
```

And for the nosecone:

```

def evaluate_center_of_pressure(self):
    """Calculates and returns the center of pressure of the nose cone in
    local coordinates. The center of pressure position is saved and stored
    as a tuple. Local coordinate origin is found at the tip of the nose
    cone.

    Returns
    -----
    self.cp : tuple
        Tuple containing cpx, cpy, cpz.
    """

    self.cpz = self.k * self.length
    self.cpy = 0
    self.cpx = 0
    self.cp = (self.cpx, self.cpy, self.cpz)
    return self.cp

```

Where the k factor depends on the geometry of the nose cone - here's the value for a von karman nose cone geometry

```

elif value == "vonkarman":
    self.k = 0.5

    def theta(x):
        return np.arccos(1 - 2 * max(min(x / self.length, 1), 0))

    self.y_nosecone = Function(
        lambda x: self.base_radius
        * (theta(x) - np.sin(2 * theta(x)) / 2) ** (0.5)
        / (np.pi**0.5)
    )

```

In any case here is the formulation of the center of pressure for an airbrake instance of an aerodynamic surface



```

def evaluate_center_of_pressure(self):
    """Evaluates the center of pressure of the aerodynamic surface in local
    coordinates.

    For air brakes, all components of the center of pressure position are
    0.

    Returns
    -----
    None
    """
    self.cpx = 0
    self.cpy = 0
    self.cpz = 0
    self.cp = (self.cpx, self.cpy, self.cpz)

```

And it becomes clear that Rocket Py also operates under the assumption that symmetrical do not effect the CP.

**Design and Analysis of a High Powered Rocket Air-brake System. 'Space Cowboys'**  
**Spaceport 2019. Mississippi State University -** <https://arc.aiaa.org/doi/epdf/10.2514/6.2024-85628>

<https://arc.aiaa.org/doi/pdf/10.2514/6.2024-85628>

**Design and Testing of an Airbrake System for a Sounding Rocket -**  
<https://arc.aiaa.org/doi/epdf/10.2514/6.2025-98650>

**How to Design Airbrakes in a Sounding Rocket -** <https://www.apogeerockets.com/Peak-of-Flight/Newsletter599>

[https://www.grc.nasa.gov/www/k-12/VirtualAero/BottleRocket/airplane/shaped.html?utm\\_source=chatgpt.com](https://www.grc.nasa.gov/www/k-12/VirtualAero/BottleRocket/airplane/shaped.html?utm_source=chatgpt.com)

Lew, T., Lyck, F., & Müller, G. (2019). **Chance-Constrained Optimal Altitude Control of a Rocket.** In 8th European Conference for Aeronautics and Aerospace Sciences (EUCASS) (pp. 1-15). DOI: 10.13009/EUCASS2019-388

FLUID-DYNAMIC DRAG theoretical, experimental and statistical information presented by  
 Dr.-Ing. S. F. HOERNER 1965



